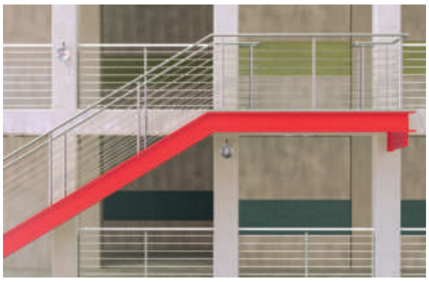
Injection mortar FIS V Plus

The powerful universal mortar for concrete and masonry



Steel constructions

Rescue ladders

Applications

Injection mortar for use with:
· Threaded rods FIS A, see page 128
· Internal threaded anchor RG MI, see page 142
· Rebar anchor FRA, see page 170
· Concrete steel bars, see page 177
· Injection anchor sleeves FIS H, see page 149
· Aerated concrete centring sleeve PBZ, see page 194
· Remedial wall tie VBS 8, see page 179
· Weather facing reconstruction system FWS II, see page 181

Advantages

· The FIS V Plus injection mortar has numerous system approvals, such as in cracked and non-cracked concrete, masonry and for special applications.
· The ETA assessment for a service life of 100 years offers permanent safety for all applications.
· The approved use in water-filled drill holes enables a wide range of applications, even under harsh environmental conditions.
· FIS VW Plus High Speed has a significantly shorter curing time than FIS V Plus, thus also ensuring swift work progress even at low temperatures.
· Due to the possible installation temperature of -10° to 40°C the universal mortar can be applied all year long.
· FIS VS Plus Low Speed with extended gelling time prevents premature curing of the mortar at higher temperatures and is ideally suited to large drill hole depths.
· The extensive range of accessories is ideally suited to the FIS V Plus injection mortar family, increases the great flexibility of the system and thus allows for a broad range of applications.

Certificates

ETA CE

ICC ES ESR-2786 See ICC-ES Evaluation Report at www.icc-es.org

ETA-20/0603, for concrete

ETA-20/0728, for post-installed rebar connections

ETA-20/0729, for masonry

SEISMIC

Fire resistance classification R120

100 Years
Service life

Building materials

Approved for anchorings in:
· Concrete C20/25 to C50/60, cracked and non-cracked
· Hollow blocks made from lightweight concrete
· Hollow blocks made from concrete
· Vertically perforated brick
· Perforated sand-lime brick
· Solid sand-lime brick
· Aerated concrete
· Solid brick

Approved for:
· Rebar connections
· Remedial wall tie VBS 8
· Weather facing reconstruction system FWS II
· Stand-off installation TherMax

Functioning

· The FIS V Plus is a 2-component injection mortar based on vinyl ester hybrid.
· Resin and hardener are stored in two separate chambers and are not mixed and activated until extrusion through the static mixer.
· The injection cartridges are quick and easy to use with the fischer dispensers.
· Partially used cartridges can be reused, simply by changing the static mixer.

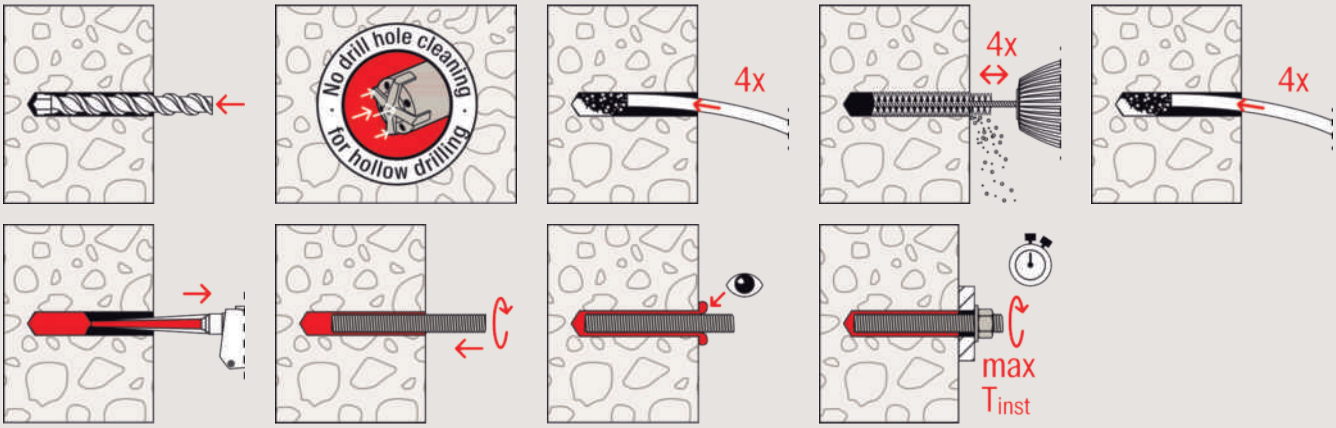
For use with

Anchors & sleeves page 71

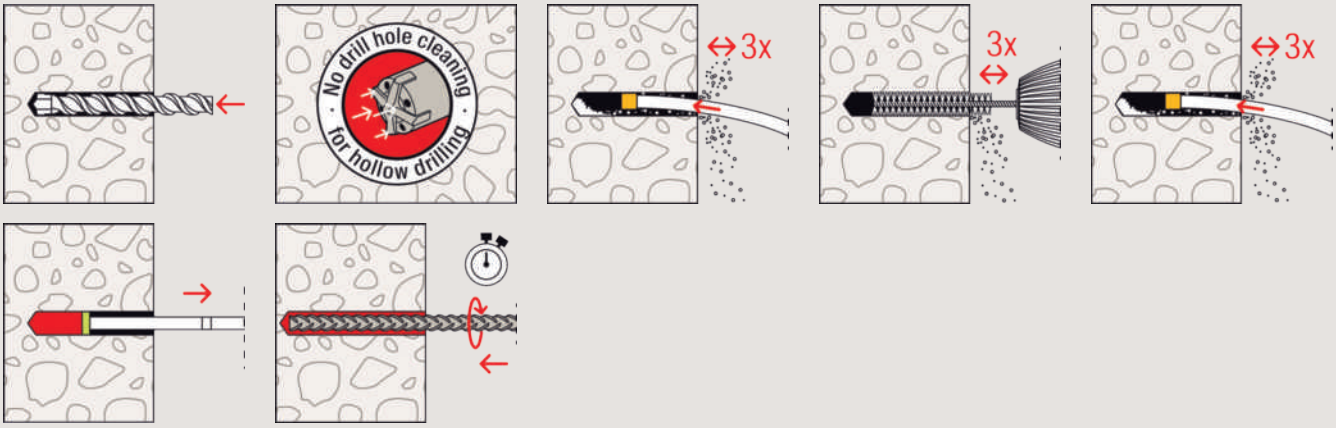
Dispenser page 87

Accessories page 89

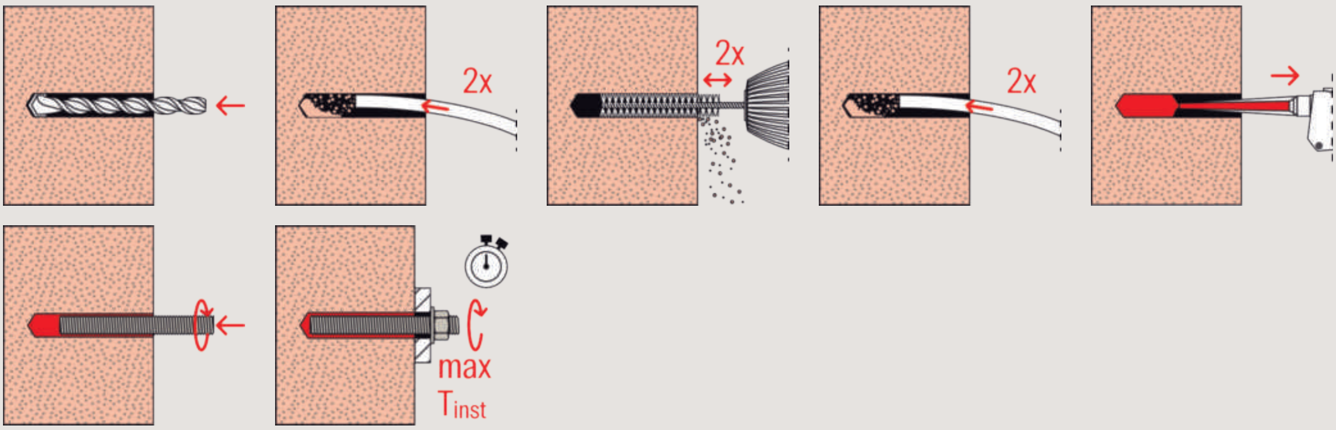
Installation in concrete with FIS V Plus and FIS A / RG M



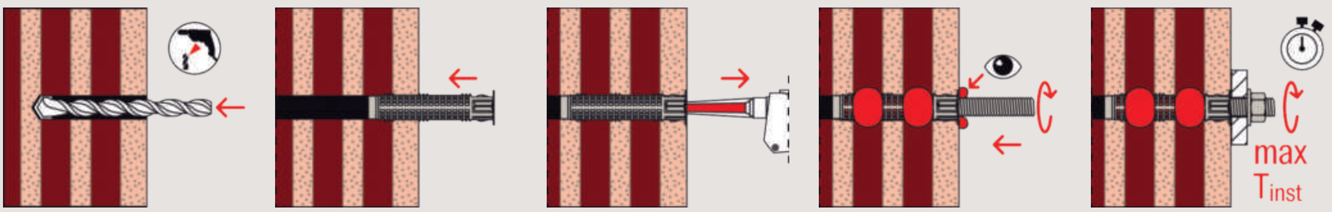
Installation Installation in concrete with FIS V Plus and FIS V in hammer drilled holes in concrete with FIS V Plus in hammer drilled holes



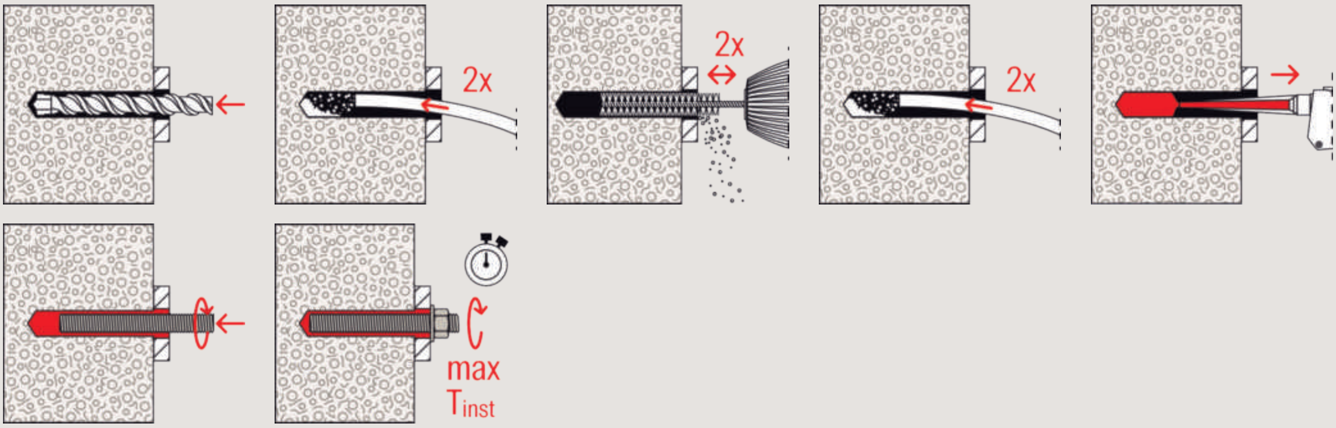
Installation in solid brick with FIS V Plus and FIS A



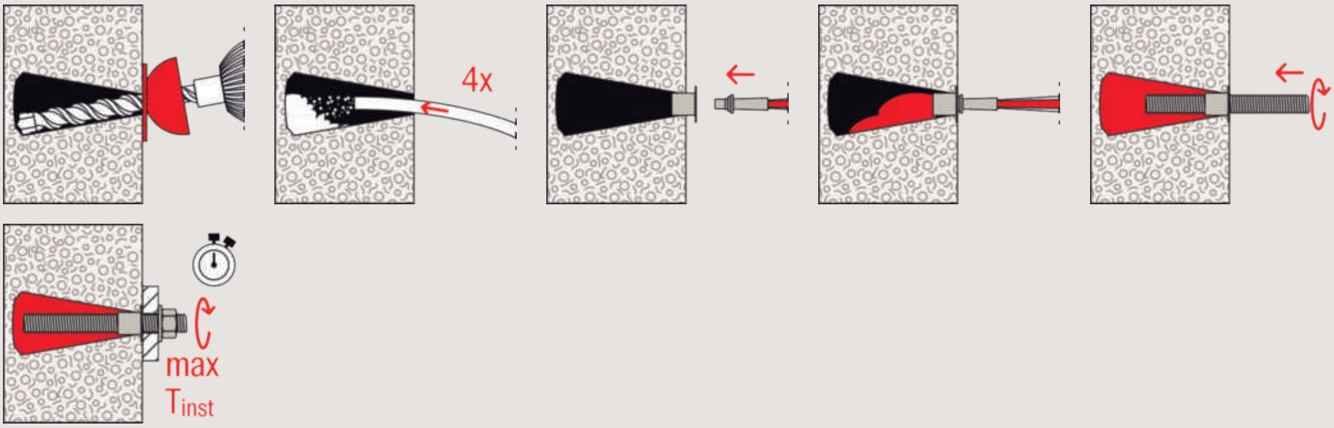
Installation in hollow blocks with FIS V Plus and FIS HK + FIS A



Installation in aerated concrete with FIS V Plus and FIS A / RG M



Installation in undercut drill hole in aerated concrete with FIS V Plus and FIS A / RG M



Technical data

Injection mortar FIS V Plus

FIS V Plus 360 S

Item	Item No.	Approval DIBt	Approval ETA	Approval ICC	Languages on the cartridge	Contents	Sales unit [pcs]
FIS V Plus 360 S (AR,ZH,EN)	558747	●	●	●	AR, ZH, EN	1 cartridge 360 ml, 2 x FIS MR Plus	6

Technical data

Injection mortar FIS V Plus

FIS V Plus 360 S BT

Item	Item No.	Approval DIBt	Approval ETA	Approval ICC	Languages on the cartridge	Contents	Sales unit [pcs]
FIS V Plus 360 S (AR,ZH,EN) BT	558751	●	●	●	AR, ZH, EN	20 cartridges 360 ml, 20 static mixer FIS MR Plus	1

Curing times

FIS VW Plus High Speed FIS V Plus FIS VS Plus Low Speed										
Temperature at anchoring base	Maximum processing time t_{work}			Minimum curing time t_{cure} [1)]						
	FIS VW Plus High Speed	FIS V Plus	FIS VS Plus Low Speed	FIS VW Plus High Speed		FIS V Plus		FIS VS Plus Low Speed		
[°C]	[min.]	[min.]	[min.]	[hrs.]	[min.]	[hrs.]	[min.]	[hrs.]	[min.]
-10 – -5 [2)]	–	–	–	12	–	–	–	–	–
> -5 – 0 [2)]	5	–	–	3	–	24	–	–	–
> 0 – +5 [2)]	5	13	–	3	–	3	–	6	–
> +5 – +10	3	9	20	–	50	–	90	3	–
> +10 – +20	1	5	10	–	50	–	60	2	–
> +20 – +30	–	4	6	–	–	–	45	–	60
> +30 – +40	–	2	4	–	–	–	35	–	30

1) In wet concrete or water filled holes the curing times must be doubled.
2) Minimal cartridge temperature +5 °C

Loads

Injection system FIS V Plus with threaded rod FIS A resp. RG M

Permissible loads of a single anchor[1) 2)] in normal concrete of strength class C20/25.
For the design the complete current assessment ETA-20/0603 has to be considered.

					Cracked concrete Permissible tension (N_{perm}) and shear loads (V_{perm}); minimum spacing (s_{min}) and edge distances (c_{min}) with reduced loads				Non-cracked concrete Permissible tension (N_{perm}) and shear loads (V_{perm}); minimum spacing (s_{min}) and edge distances (c_{min}) with reduced loads			
Type	Material / surface[3)]	Effective anchorage depth h_{ef} [mm]	Minimum member thickness h_{min} [mm]	Maximum instal-lation torque $T_{inst, max}$ [Nm]	N_{perm} [4)] [kN]	V_{perm} [4)] [kN]	s_{min} [4)] [mm]	c_{min} [4)] [mm]	N_{perm} [4)] [kN]	V_{perm} [4)] [kN]	s_{min} [4)] [mm]	c_{min} [4)] [mm]
FIS A M 8	5.8	60	100	10	3.9	6.3	40	40	9.0	6.3	40	40
	5.8	80	110	10	5.3	6.3	40	40	9.0	6.3	40	40
	5.8	160	190	10	9.0	6.3	40	40	9.0	6.3	40	40
	R-70	60	100	10	3.9	6.0	40	40	9.9	6.0	40	40
	R-70	80	110	10	5.3	6.0	40	40	9.9	6.0	40	40
	R-70	160	190	10	9.9	6.0	40	40	9.9	6.0	40	40
FIS A M 10	5.8	60	100	20	5.4	9.7	45	45	10.9	9.7	45	45
	5.8	90	120	20	8.1	9.7	45	45	13.8	9.7	45	45
	5.8	200	230	20	13.8	9.7	45	45	13.8	9.7	45	45
	R-70	60	100	20	5.4	9.2	45	45	10.9	9.2	45	45
	R-70	90	120	20	8.1	9.2	45	45	15.7	9.2	45	45
	R-70	200	230	20	15.7	9.2	45	45	15.7	9.2	45	45
FIS A M 12	5.8	70	100	40	8.2	14.3	55	45	13.7	14.3	55	45
	5.8	110	140	40	12.8	14.3	55	45	20.5	14.3	55	45
	5.8	240	270	40	20.5	14.3	55	45	20.5	14.3	55	45
	R-70	70	100	40	8.2	13.7	55	45	13.7	13.7	55	45
	R-70	110	140	40	12.8	13.7	55	45	22.5	13.7	55	45
	R-70	240	270	40	22.5	13.7	55	45	22.5	13.7	55	45
FIS A M 16	5.8	80	120	60	11.5	23.0	65	50	16.8	26.9	65	50
	5.8	125	170	60	18.0	26.9	65	50	32.7	26.9	65	50
	5.8	320	360	60	37.6	26.9	65	50	37.6	26.9	65	50
	R-70	80	120	60	11.5	23.0	65	50	16.8	25.2	65	50
	R-70	125	170	60	18.0	25.2	65	50	32.7	25.2	65	50
	R-70	320	360	60	42.0	25.2	65	50	42.0	25.2	65	50
FIS A M 20	5.8	90	140	120	14.0	28.0	85	55	20.0	40.0	85	55
	5.8	170	220	120	28.0	42.3	85	55	51.9	42.3	85	55
	5.8	400	450	120	58.6	42.3	85	55	58.6	42.3	85	55
	R-70	90	140	120	14.0	28.0	85	55	20.0	39.4	85	55
	R-70	170	220	120	28.0	39.4	85	55	51.9	39.4	85	55
	R-70	400	450	120	65.7	39.4	85	55	65.7	39.4	85	55
FIS A M 24	5.8	96	160	150	15.4	30.8	105	60	22.0	44.1	105	60
	5.8	210	270	150	37.7	60.6	105	60	71.3	60.6	105	60
	5.8	480	540	150	84.3	60.6	105	60	84.3	60.6	105	60
	R-70	96	160	150	15.4	30.8	105	60	22.0	44.1	105	60
	R-70	210	270	150	37.7	56.8	105	60	71.3	56.8	105	60
	R-70	480	540	150	86.2	56.8	105	60	94.3	56.8	105	60
FIS A M 30	5.8	120	190	300	21.6	43.1	140	80	30.8	61.6	140	80
	5.8	280	350	300	56.5	96.0	140	80	109.8	96.0	140	80
	5.8	600	670	300	121.2	96.0	140	80	133.8	96.0	140	80
	R-70	120	190	300	21.6	43.1	140	80	30.8	61.6	140	80
	R-70	280	350	300	56.5	90.2	140	80	109.8	90.2	140	80
	R-70	600	670	300	121.2	90.2	140	80	150.1	90.2	140	80

1) Design according to EN 1992-4:2018 (for static resp. quasi-static loads). The partial safety factors for material resistance as regulated in the ETA as well as a partial safety factor for load actions of γ_L = 1.4 are considered. As a single anchor counts e.g. an anchor with a spacing s ≥ 3 x h_{ef} and an edge distance c ≥ 1.5 x h_{ef}. Accurate data see ETA.
2) The specified loads are valid for anchorages in dry and damp concrete. For temperatures in the anchoring substrate up to 50 °C (resp. short term up to 80 °C). Drill hole cleaning as per specification in the ETA. The factor Ψ_{sus} for sustained load was taken into account with 1.0.
3) Further steel grades, versions and technical data see ETA, e.g. for dry internal conditions, galvanised steel (gvz); for damp interiors and for outdoor use, stainless steel (R).
4) In the case of combinations of tension and shear loads, bending moments with reduced or minimum spacing and edge distances (anchor groups), the design must be carried out in accordance with the provisions of the complete ETA and the provisions of the EN 1992-4:2018. We recommend using our anchor design software C-FIX.

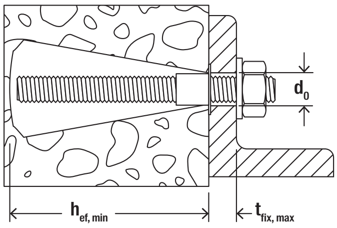
Loads

Injection system FIS V Plus with threaded rod FIS A in solid and perforated masonry

Permissible loads[1) 2)] for a single anchor in masonry for pre-positioned installation.
For the design the complete current assessment ETA-20/0729 has to be considered.

Type	Compressive brick strength f_b [N/mm²]	Brick raw density ρ [kg/dm³]	Minimum brick dimensions[3)] (L x W x H) [mm]	Effective anchorage depth h_{ef} [mm]	Minimum member thickness h_{min} [mm]	Maximum installation torque $T_{inst,max}$ [Nm]	Permissible tensile load[4)] N_{perm} [kN]	Permissible shear load[4)] V_{perm} [kN]	Minimum-spacing[5)] $s_{min\,\parallel}$ / $s_{min\perp}$ [mm]	Characteristic resp. minimum edge distance[5)] $c_{cr} = c_{min}$ [mm]
Solid brick Mz, NF, acc. to EN 771-1										
M6	≥ 12	≥ 1.8	240 x 115 x 71	≥ 50	115	4	1.14	0.71	240 / 75	100
M8	≥ 12	≥ 1.8	240 x 115 x 71	≥ 50	115	10	1.14	0.71	240 / 75	100
M10	≥ 12	≥ 1.8	240 x 115 x 71	80	115	10	1.42	1.14	240 / 75	100
M10	≥ 12	≥ 1.8	240 x 115 x 71	200	240	10	3.43	2.43	240 / 75	100
M12	≥ 12	≥ 1.8	240 x 115 x 71	80	115	10	1.57	1.14	240 / 75	100
M12	≥ 12	≥ 1.8	240 x 115 x 71	200	240	10	2.29	3.28	240 / 75	100
Solid sand-lime brick KS, acc. to EN 771-2										
M6	≥ 12	≥ 1.8	240 x 115 x 71	50	115	3	1.14	0.42	80 / 150	60
M6	≥ 12	≥ 1.8	240 x 115 x 71	100	115	3	1.57	0.89	80 / 300	60
M8	≥ 12	≥ 1.8	240 x 115 x 71	50	115	5	1.14	0.42	80 / 150	60
M8	≥ 12	≥ 1.8	240 x 115 x 71	100	115	5	2.29	0.89	80 / 300	60
M10	≥ 12	≥ 1.8	240 x 115 x 71	100	115	15	1.57	0.57	80 / 300	60
M10	≥ 12	≥ 1.8	240 x 115 x 71	200	240	15	3.42	0.57	80 / 600	60
M12	≥ 12	≥ 1.8	240 x 115 x 71	100	115	15	1.28	0.57	80 / 300	60
M12	≥ 12	≥ 1.8	240 x 115 x 71	200	240	15	3.42	0.57	80 / 600	60
M16	≥ 12	≥ 1.8	240 x 115 x 71	100	115	25	1.57	0.57	80 / 300	60
M16	≥ 12	≥ 1.8	240 x 115 x 71	200	240	25	3.42	0.57	80 / 600	60
Vertically perforated brick Hlz, acc. to EN 771-1[3)]										
M6 / M8 with FIS H 12 x 85 K	≥ 12	≥ 1.0	370 x 240 x 237	85	240	2	0.34	0.43	100 / 100	100
M8 / M10 with FIS H 16 x 130 K	≥ 12	≥ 1.0	370 x 240 x 237	130	240	2	0.86	0.57	100 / 100	100
M12 / M16 with FIS H 20 x 130 K	≥ 12	≥ 1.0	370 x 240 x 237	130	240	2	1.14	0.57	100 / 100	100
Perforated sand-lime brick KSL, acc. to EN 771-2[3)]										
M6 / M8 with FIS H 12 x 85 K	≥ 12	≥ 1.4	240 x 175 x 113	85	175	2	0.71	0.71	100 / 115	60
M8 / M10 with FIS H 16 x 130 K	≥ 12	≥ 1.4	240 x 175 x 113	130	175	2	1.00	1.29	100 / 115	80
M12 / M16 with FIS H 20 x 85 K	≥ 12	≥ 1.4	240 x 175 x 113	85	175	2	1.00	1.14	100 / 115	80
Lightweight concrete hollow block Hbl, acc. EN 771-3[3)]										
M6 / M8 with FIS H 12 x 85 K	≥ 2	≥ 1.0	362 x 240 x 240	85	240	2	0.43	0.26	100 / 240	60
M6 / M8 with FIS H 12 x 85 K	≥ 4	≥ 1.0	362 x 240 x 240	85	240	2	0.86	0.57	100 / 240	60
M8 / M10 with FIS H 16 x 85 K	≥ 2	≥ 1.0	362 x 240 x 240	85	240	2	0.43	0.26	100 / 240	60
M8 / M10 with FIS H 16 x 85 K	≥ 4	≥ 1.0	362 x 240 x 240	85	240	2	0.86	0.57	100 / 240	60
M12 / M16 with FIS H 20 x 200 K	≥ 2	≥ 1.0	362 x 240 x 240	200	240	2	0.71	0.26	100 / 240	60
M12 / M16 with FIS H 20 x 200 K	≥ 4	≥ 1.0	362 x 240 x 240	200	240	2	1.57	0.57	100 / 240	60
Aerated concrete acc. to EN 771-4[6)]										
M8	≥ 2	≥ 0.35	-	100	130	1	0.54	0.43	250 / 250	100
M8	≥ 4	≥ 0.50	-	200	230	8	1.07	0.71	80 / 80	100
M10	≥ 2	≥ 0.35	-	100	130	2	0.54	0.43	250 / 250	100
M10	≥ 4	≥ 0.50	-	200	230	12	1.79	0.71	80 / 80	100
M12	≥ 2	≥ 0.35	-	100	130	2	0.71	0.54	250 / 250	100
M12	≥ 4	≥ 0.50	-	200	230	16	1.79	0.71	80 / 80	100
M16	≥ 2	≥ 0.35	-	100	130	2	0.71	0.43	250 / 250	100
M16	≥ 4	≥ 0.50	-	200	230	20	1.79	0.71	80 / 80	100

1) The required partial safety factors for material resistance as well as a partial safety factor for load actions of γ_L = 1.4 are considered. Load values are valid for zinc-plated steel, stainless steel R and highly corrosion-resistant steel HCR. In perforated bricks and hollow blocks threaded rod FIS A in combination with anchor sleeve FIS H K.
2) The given loads are valid for installation and use of fixations in dry masonry - use category d/d - for temperatures in the substrate up to 50 °C (resp. short term up to 80 °C) and drill hole cleaning according to assessment. The given brick types in combination with the permissible loads are an extract of the assessment.
3) More information about, e.g. hole patterns, assortment of anchor sleeves FIS H K see assessment.
4) In the case of combinations of tensile and shear loads, bending moments and reduced edge and axial spacings (anchor groups), the design must be carried out in accordance with the provisions of the complete assessment.
5) Minimum feasible spacing resp. edge distance. Details as well as to the distances to joints see assessment.
6) Cylindrical drill hole.



Technical data in undercut drill hole in aerated concrete

2

Threaded rod FIS A

FIS A

	Zinc plated, steel grade 5.8 Item No.	Zinc plated, steel grade 8.8 Item No.	Stainless steel Item No.	Approval	Drill hole diameter in aerated concrete in undercut drill hole [mm]	Min. / max. anchorage depth in aerated concrete in undercut drill hole h_{ef} [mm]	Min. / max. usable length in aerated concrete t_{fix} [mm]	Filling quantity for min. / max. anchorage depth in aerated concrete [scale units]	Sales unit [pcs]
Item	gvz	gvz	R	ETA					
FIS A M 8 x 90	090274	519390	090440	●	14	75 / 95	4 / -	15 / 20	10
FIS A M 8 x 110	090275	519391	090441	●	14	75 / 95	24 / 4	15 / 20	10
FIS A M 8 x 130	090276	519392	090442	●	14	75 / 95	44 / 24	15 / 20	10
FIS A M 8 x 140	–	553763	–	●	14	75 / 90	55 / 40	15 / 20	10
FIS A M 8 x 175	090277	519393	090443	●	14	75 / 95	89 / 69	15 / 20	10
FIS A M 10 x 110	090278	–	090444	●	14	75 / 95	22 / 2	15 / 20	10
FIS A M 10 x 130	090279	524170	090447	●	14	75 / 95	42 / 22	15 / 20	10
FIS A M 10 x 150	090281	517935	090448	●	14	75 / 95	62 / 42	15 / 20	10
FIS A M 10 x 170	044969	519395	044973	●	14	75 / 95	82 / 62	15 / 20	10
FIS A M 10 x 190	–	517936	–	●	14	75 / 95	102 / 82	15 / 20	10
FIS A M 10 x 200	090282	519396	090449	●	14	75 / 95	112 / 92	15 / 20	10
FIS A M 12 x 120	044971	519397	044974	●	14	75 / 95	29 / 9	15 / 20	10

Approval body for construction products and types of construction
Bautechnisches Prüfamt
An institution established by the Federal and Laender Governments

Designated according to Article 29 of Regulation (EU) No 305/2011 and member of EOTA (European Organisation for Technical Assessment)

European Technical Assessment ETA-20/0728 of 16 December 2022

English translation prepared by DIBt - Original version in German language

General Part

Technical Assessment Body issuing the European Technical Assessment:	Deutsches Institut für Bautechnik
Trade name of the construction product	Rebar connection with injection system FIS V Plus
Product family to which the construction product belongs	Systems for post-installed rebar connections with mortar
Manufacturer	fischerwerke GmbH & Co. KG Otto-Hahn-Straße 15 79211 Denzlingen DEUTSCHLAND
Manufacturing plant	fischerwerke
This European Technical Assessment contains	24 pages including 3 annexes which form an integral part of this assessment
This European Technical Assessment is issued in accordance with Regulation (EU) No 305/2011, on the basis of	EAD 330087-01-0601, Edition 06/2021
This version replaces	ETA-20/0728 issued on 13 November 2020

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Specific Part

1 Technical description of the product

The subject of this European Technical Assessment is the post-installed connection, by anchoring or overlap connection joint, of reinforcing bars (rebars) in existing structures made of normal weight concrete, using the "Rebar connection with injection system FIS V Plus" in accordance with the regulations for reinforced concrete construction.

Reinforcing bars made of steel with a diameter ϕ from 8 to 28 mm or the fischer rebar anchor FRA or FRA HCR of sizes M12 to M24 according to Annex A and injection mortar FIS V Plus or FIS V Plus Low Speed are used for rebar connections. The rebar is placed into a drilled hole filled with injection mortar and is anchored via the bond between rebar, injection mortar and concrete.

The product description is given in Annex A.

2 Specification of the intended use in accordance with the applicable European assessment Document

The performances given in Section 3 are only valid if the rebar connection is used in compliance with the specifications and conditions given in Annex B.

The verifications and assessment methods on which this European Technical Assessment is based lead to the assumption of a working life of the rebar connections of at least 50 years. The indications given on the working life cannot be interpreted as a guarantee given by the producer, but are to be regarded only as a means for choosing the right products in relation to the expected economically reasonable working life of the works.

3 Performance of the product and references to the methods used for its assessment

3.1 Mechanical resistance and stability (BWR 1)

Essential characteristic	Performance
Characteristic resistance under static and quasi-static loading	See Annex C 1 and C2
Characteristic resistance under seismic loading	No performance assessed

3.2 Safety in case of fire (BWR 2)

Essential characteristic	Performance
Reaction to fire	Class A1
Resistance to fire	See Annex C 2 and C 3

4 Assessment and verification of constancy of performance (AVCP) system applied, with reference to its legal base

In accordance with European Assessment Document EAD No. 330087-01-0601, the applicable European legal act is: [96/582/EC].

The system to be applied is: 1

5 Technical details necessary for the implementation of the AVCP system, as provided for in the applicable European Assessment Document

Technical details necessary for the implementation of the AVCP system are laid down in the control plan deposited with Deutsches Institut für Bautechnik.

Issued in Berlin on 16 December 2022 by Deutsches Institut für Bautechnik

Dipl.-Ing. Beatrix Wittstock
Head of Section

beglaubigt:
Baderschneider

Installation conditions and application examples reinforcing bars, part 1

Figure A1.1:
Overlap joint with existing reinforcement for rebar connections of slabs and beams

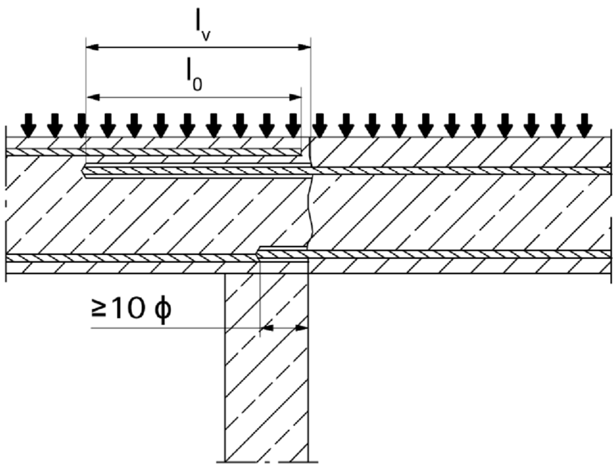


Figure A1.2:
Overlap joint with existing reinforcement at a foundation of a column or wall where the rebars are stressed

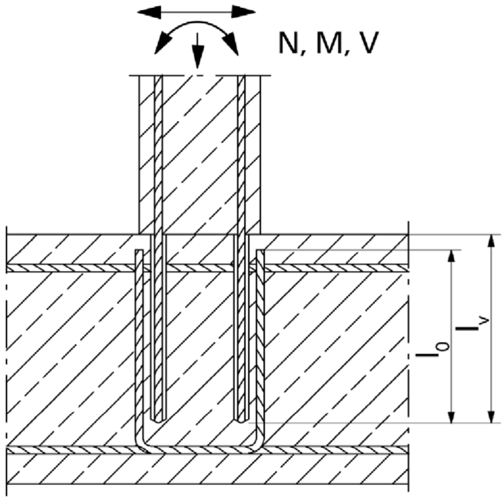
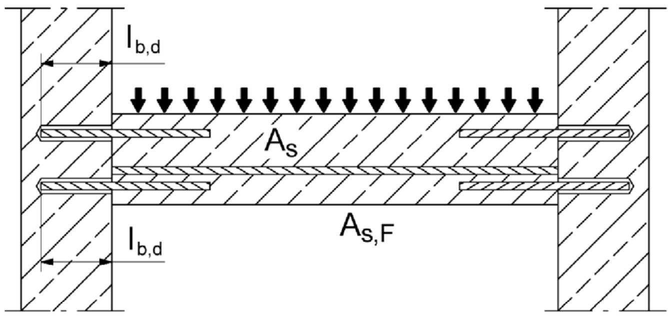


Figure A1.3:
End anchoring of slabs or beams (e.g. designed as simply supported)



Figures not to scale

Rebar connection with injection system FIS V Plus	Annex A 1
Product description Installation conditions and application examples reinforcing bars, part 1	

Installation conditions and application examples reinforcing bars, part 2

Figure A2.1:
Rebar connection for stressed primarily in compression

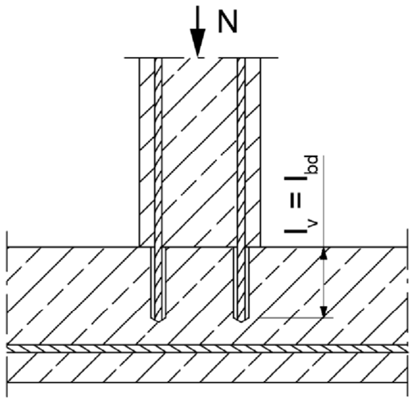
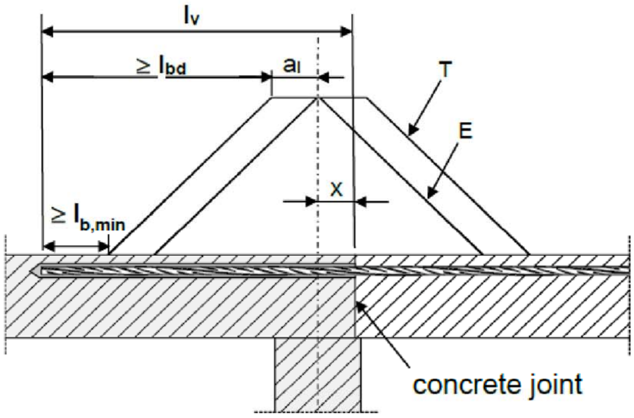


Figure A2.2:
Anchoring of reinforcement to cover the enveloped line of acting tensile force in the bending member



(only post-installed rebar is plotted)

Key to Figure

- T Acting tensile force
- E Envelope of M_{ed} / z + N_{ed} (see EN 1992-1-1:2011)
- x Distance between the theoretical point of support and concrete joint

Note to **figure A1.1** to **A1.3** and **figure A2.1** to **A2.2**

In the figures no traverse reinforcement is plotted, the transverse reinforcement as required by EN 1992-1-1:2011 shall be present.

The shear transfer between old and new concrete shall be designed according to EN 1992-1-1:2011
Preparation of joints according to **Annex B 3** of this document

Figures not to scale

Rebar connection with injection system FIS V Plus	**Annex A 2**
Product description Installation conditions and application examples reinforcing bars, part 2	

Installation conditions and application examples fischer rebar anchor FRA

Figure A3.1:
Lap to a foundation of a column under bending.

1. Shear lug (or fastener loaded in shear)
2. fischer rebar anchor FRA (tension only)
3. Existing stirrup / reinforcement for overlap (lap splice)
4. Slotted hole

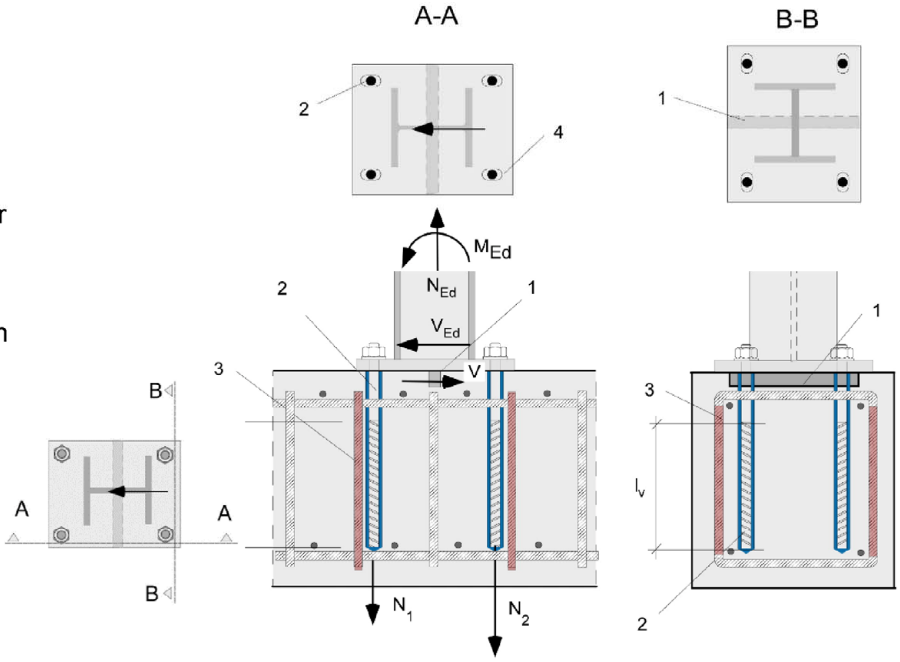
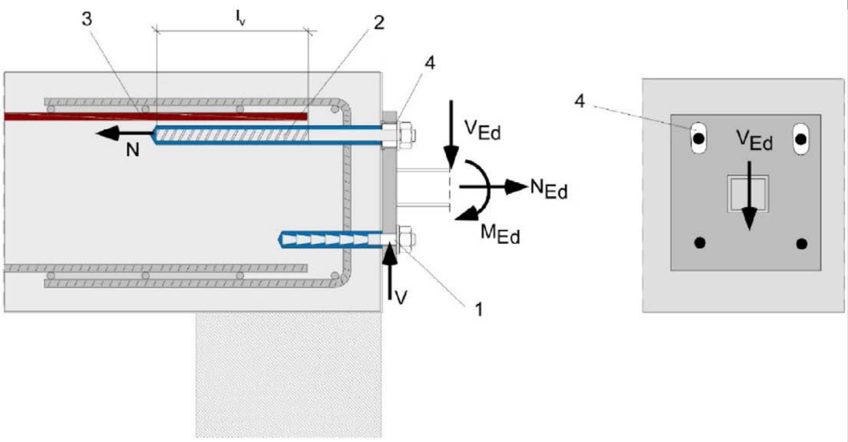


Figure A3.2:
Lap of the anchoring of guardrail posts or anchoring of cantilevered building components.
In the anchor plate, the drill holes for the fischer rebar anchors FRA have to be designed as slotted holes with axial direction to the shear force.

1. Fastener for shear load transfer
2. fischer rebar anchor FRA (tension only)
3. Existing stirrup / reinforcement for overlap (lap splice)
4. Slotted hole



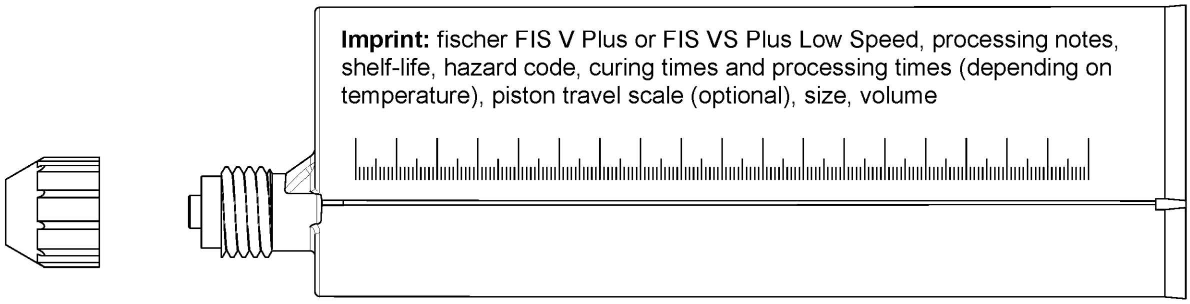
The required transverse reinforcement acc. to EN 1992-1-1:2011 is not shown in the figures. **The fischer rebar anchor FRA may be only used for axial tensile force.** The tensile force must be transferred by lap to the existing reinforcement of the building. The transfer of the shear force has to be ensured by suitable measure, e.g. by means of shear force or anchors with European Technical Assessment (ETA).

Figures not to scale

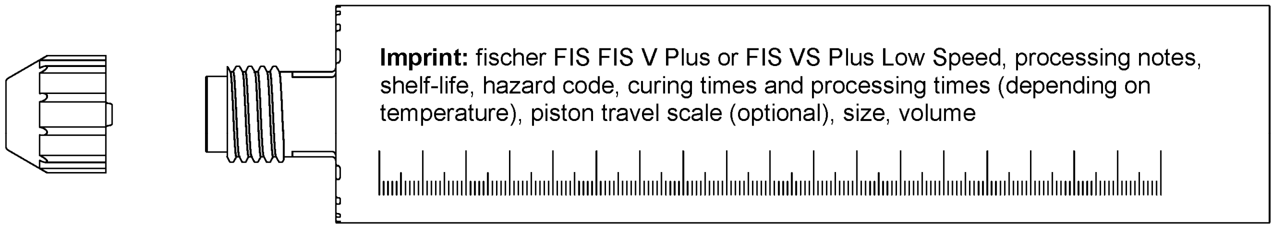
Rebar connection with injection system FIS V Plus	Annex A 3
Product description Installation conditions and application examples fischer rebar anchors FRA	

Overview system components

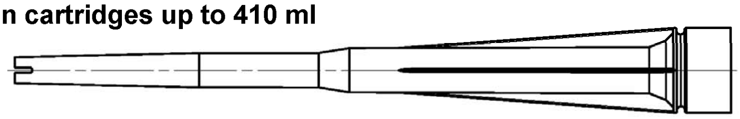
Injection cartridge (shuttle cartridge) FIS V Plus with sealing cap; Sizes: 360 ml, 825 ml



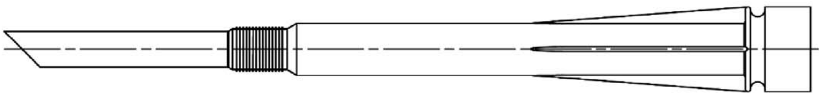
Injection cartridge (coaxial cartridge) FIS V Plus with sealing cap; Sizes: 300 ml, 380 ml, 400 ml, 410 ml



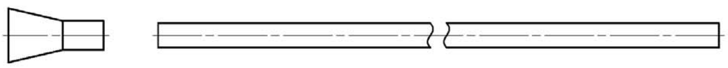
Static mixer FIS MR Plus for injection cartridges up to 410 ml



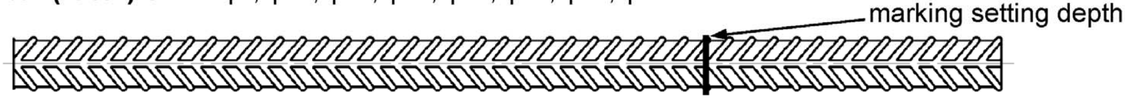
Static mixer FIS JMR for injection cartridges 825 ml



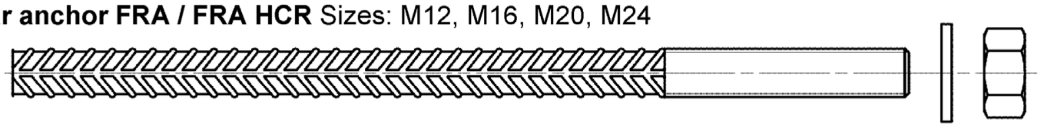
Injection adapter and extension tube Ø 9 for static mixer FIS MR Plus;
Injection adapter and extension tube Ø 9 or Ø 15 for static mixer FIS JMR



Reinforcing bar (rebar) Sizes: ϕ8, ϕ10, ϕ12, ϕ14, ϕ16, ϕ20, ϕ25, ϕ28

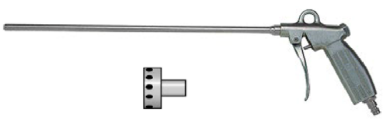


fischer rebar anchor FRA / FRA HCR Sizes: M12, M16, M20, M24



Blow out pump AB G

Compressed-air cleaning tool ABP with fischer compressed-air nozzle

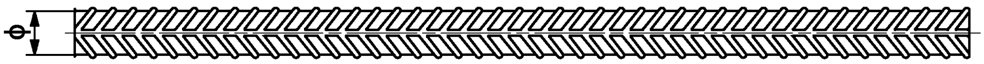


Figures not to scale

Rebar connection with injection system FIS V Plus	**Annex A 4**
Product description Overview system components; Injection mortar, static mixer, injection adapter, reinforcing bar, fischer rebar anchor FRA, cleaning tools	

Properties of reinforcing bars (rebar)

Figure A5.1:



- The minimum value of related rib area $f_{R,min}$ according to EN 1992-1-1:2011
- The maximum outer rebar diameter over the ribs shall be:
  - The nominal diameter of the bar with rib $\phi + 2 \cdot h$ ($h \leq 0{,}07 \cdot \phi$)
  - (ϕ: Nominal diameter of the bar; h_{rib} = rib height of the bar)

Table A5.1: Installation conditions for rebars

Nominal diameter of the bar	**ϕ**		**8[1)]**		**10[1)]**		**12[1)]**		**14**	**16**	**20**	**25[1)]**		**28**
Nominal drill hole diameter	d_0	[mm]	10	12	12	14	14	16	18	20	25	30	35	35
Drill hole depth	h_0	[mm]	$h_0 = l_v$											
Effective embedment depth	l_v	[mm]	acc. to static calculation											
Minimum thickness of concrete member	h_{min}	[mm]	$l_v + 30$ (≥ 100)						$l_v + 2d_0$					

1) Both drill hole diameters can be used

Table A5.2: Materials of rebars

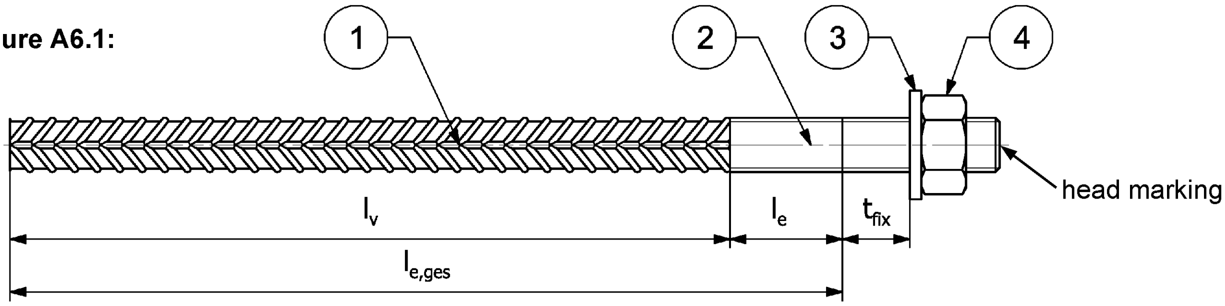
Designation	**Reinforcing bar (rebar)**
Reinforcing bar EN 1992-1-1:2011, Annex C	Bars and de-coiled rods class B or C with f_{yk} and k according to NDP or NCI of EN 1992-1-1/NA $f_{uk} = f_{tk} = k \cdot f_{yk}$

Figures not to scale

Rebar connection with injection system FIS V Plus	**Annex A 5**
Product description Properties and materials of reinforcing bars (rebar)	

Properties of fischer rebar anchors FRA

Figure A6.1:



Head marking e.g.:  FRA (for stainless steel)

 FRA HCR (for high corrosion-resistant steel)

Table A6.1: Installation conditions for fischer rebar anchors FRA

Thread diameter			M12[2)]		M16	M20	M24[2)]	
Nominal diameter	ϕ	[mm]	12		16	20	25	
Nominal drill bit diameter	d_0	[mm]	14	16	20	25	30	35
Drill hole depth ($h_0 = l_{e,ges}$)	$l_{e,ges}$	[mm]	$l_v + l_e$					
Effective embedment depth	l_v	[mm]	according to static calculation					
Distance concrete surface to welded join	l_e	[mm]	100					
Maximum Diameter of clearance hole in the fixture[1)] – Pre-positioned	d_f	[mm]	14		18	22	26	
Maximum Diameter of clearance hole in the fixture[1)] – Push through	d_f	[mm]	16	18	22	26	32	---
Minimum thickness of concrete member	h_{min}	[mm]	h_0+30		$h_0 + 2d_0$			
Maximum torque moment for attachment of the fixture	max T_{inst}	[Nm]	50		100	150	150	

[1)] For bigger clearance holes in the fixture see EN 1992-4:2018
[2)] Both drill bit diameters can be used

Table A6.2: Materials of fischer rebar anchors FRA

Part	Description	Materials: FRA Corrosion resistance class CRC III acc. to EN 1993-1-4:2006+A1:2015	Materials: FRA HCR Corrosion resistance class CRC V acc. to EN 1993-1-4: 2006+A1:2015
1	Reinforcing bar	Bars and de-coiled rods class B or C with f_{yk} and k according to NDP or NCI of EN 1992-1-1:NA; $f_{uk} = f_{tk} = k \cdot f_{yk}$; ($f_{yk}$ = 500 N/mm^2)	
2	Round bar with partial or full thread	Stainless steel, strength class 80, according to EN 10088-1:2014	Stainless steel, strength class 80, according to EN 10088-1:2014
3	Washer ISO 7089:2000	Stainless steel, according to EN 10088-1:2014	Stainless steel, according to EN 10088-1:2014
4	Hexagon nut	Stainless steel, strength class 80, acc. to EN ISO 3506-2:2020, according to EN 10088-1:2014	Stainless steel, strength class 80, acc. to EN ISO 3506-2:2020, according to EN 10088-1:2014

Figures not to scale

Rebar connection with injection system FIS V Plus

Product description
Properties and materials of fischer rebar anchors FRA

Annex A 6

Specifications of intended use part 1

Table B1.1: Overview use and performance categories

Anchorages subject to	**FIS V Plus with …**			
	Reinforcing bar 		fischer rebar anchor FRA 	
Hammer drilling or compressed air drilling with standard drill bit	all sizes			
Hammer drilling with hollow drill bit (fischer “FHD”, Heller "Duster Expert", Bosch “Speed Clean“, Hilti "TE-CD, TE-YD") 	Nominal drill bit diameter (d_0) 12 mm to 35 mm			
Use category I1 dry or wet concrete	all sizes			
Characteristic resistance under static and quasi static loading, in uncracked concrete / cracked concrete	all sizes	Tables: C1.1 C1.2 C1.3	all sizes	Tables: C1.1 C1.2 C1.3 C2.1 C2.2
Characteristic resistance under seismic loading	---[1)]		---[1)]	
Installation direction	D3 (downward and horizontal and upwards (e.g. overhead))			
Installation temperature	$T_{i,min}$ = 0 °C to $T_{i,max}$ = +40 °C			
Service temperature Temperature range	-40 °C to +80 °C		(max. short term temperature +80 °C; max long term temperature +50 °C)	
Resistance to fire	all sizes	Annex C 3	all sizes	Table C2.3

[1)] No performance assessed

Specifications of intended use part 2

Anchorages subject to:

- Static and quasi-static loading: reinforcing bar (rebar) size 8 mm to 28 mm; FRA M12 to M24
- Resistance to fire: reinforcing bar (rebar) size 8 mm to 28 mm; FRA M12 to M24

Base materials:

- Compacted reinforced or unreinforced normal weight concrete without fibres according to EN 206:2013+A1:2016
- Concrete strength classes C12/15 to C50/60 according to EN 206:2013+A1:2016
- Maximum chloride content of 0,40 % (CL 0.40) related to the cement content according to EN 206:2013+A1:2016
- Non-carbonated concrete
  Note: In case of a carbonated surface of the existing concrete structure the carbonated layer shall be removed in the area of the post-installed rebar connection with a diameter of ϕ + 60 mm prior to the installation of the new rebar. The depth of concrete to be removed shall correspond to at least the minimum concrete cover in accordance with EN 1992-1-1 :2004+AC:2010. The foregoing may be neglected if building components are new and not carbonated and if building components are in dry conditions.

Use conditions (Environmental conditions) for fischer rebar anchors FRA

- For all conditions according to EN1993-1-4:2006+A1:2015 corresponding to corrosion resistance classes to **Annex A 6 Table A6.2**.

Design:

- Fastenings are designed under the responsibility of an engineer experienced in fastenings and concrete work.
- Verifiable calculation notes and drawings are prepared taking account of the forces to be transmitted.
- Design according to EN 1992-1-1:2011; EN 1992-1-2:2011 and **Annex B 3 and B 4**.
- The actual position of the reinforcement in the existing structure shall be determined on the basis of the construction documentation and taken into account when designing.

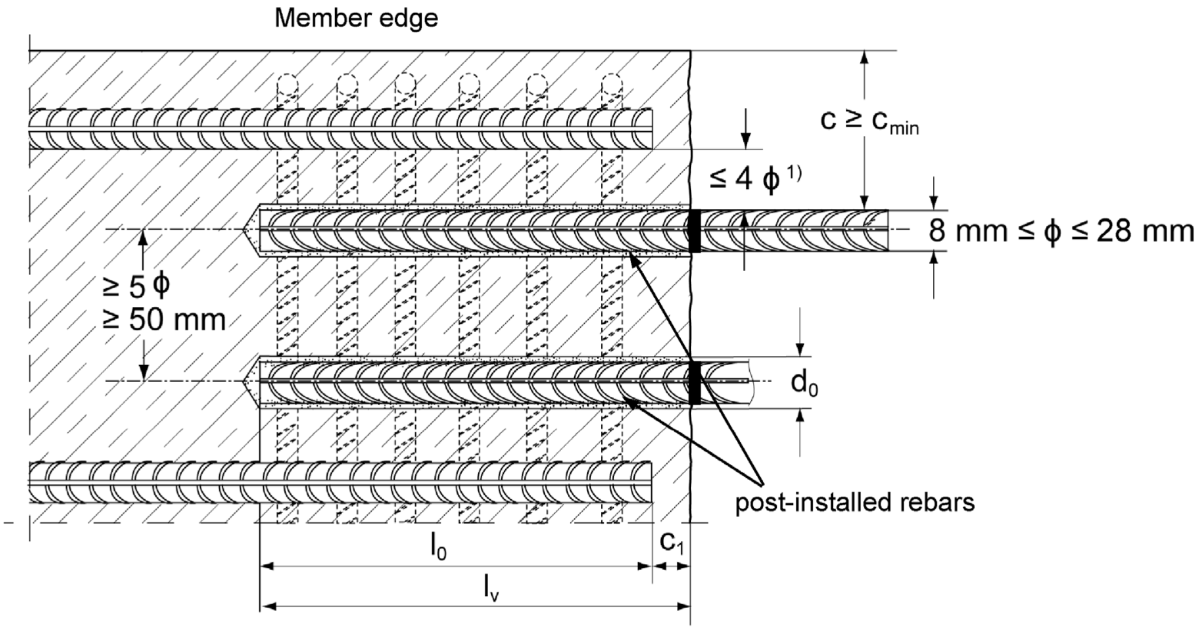
Installation:

- The installation of post-installed rebar respectively fischer rebar anchor FRA shall be done only by suitable trained installer and under Supervision on site; the conditions under which an installer may be considered as suitable trained and the conditions for Supervision on site are up to the Member States in which the installation is done.
- Check the position of the existing rebars (if the position of existing rebars is not known, it shall be determined using a rebar detector suitable for this purpose as well as on the basis of the construction documentation and then marked on the building component for the overlap joint).

General construction rules for post-installed rebars

Figure B3.1:

- Only tension forces in the axis of the rebar may be transmitted.
- The transfer of shear forces between new concrete and existing structure shall be designed additionally according to EN 1992-1-1:2011.
- The joints for concreting must be roughened to at least such an extent that aggregate protrude.



[1] If the clear distance between lapped bars exceeds 4 ϕ then the lap length shall be increased by the difference between the clear bar distance and 4 ϕ

c	concrete cover of post-installed rebar
c_1	concrete cover at end-face of existing rebar
c_{min}	minimum concrete cover according to **Table B5.1** and to EN 1992-1-1:2011, Section 4.4.1.2
ϕ	nominal diameter of reinforcing bar
l_0	lap length, according to EN 1992-1-1:2011
l_v	effective embedment depth, $\geq l_0 + c_1$
d_0	nominal drill bit diameter, see **Annex B 6**

Figures not to scale

Rebar connection with injection system FIS V Plus

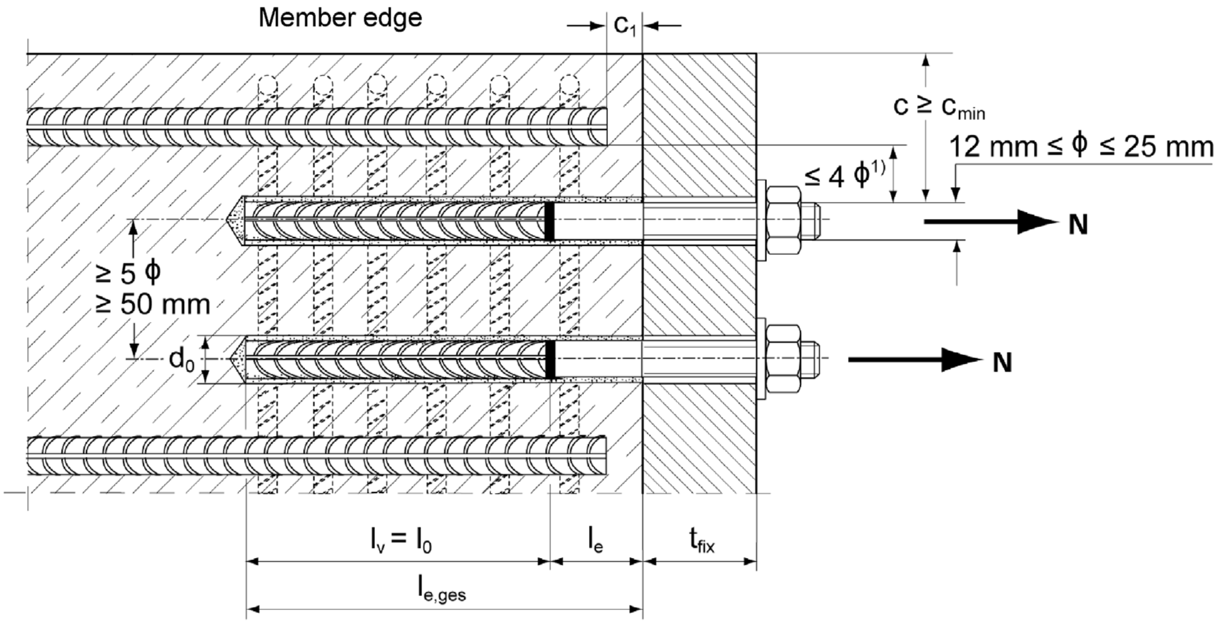
Intended use
General construction rules for post-installed rebars

Annex B 3

General construction rules for post-installed fischer rebar anchors FRA

Figure B4.1:

- Only tension forces in the axis of the fischer rebar anchor FRA may be transmitted.
- The tension force must be transferred via an overlap joint to the reinforcement in the building part.
- The transmission of the shear load shall be ensured by appropriate additional measures, e.g. by shear lugs or by anchors with a European Technical Assessment (ETA).
- In the anchor plate, the holes for the tension anchor shall be executed as slotted holes with the axis in the direction of the shear force.



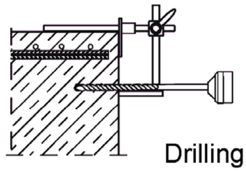
[1)] If the clear distance between lapped bars exceeds 4 ϕ then the lap length shall be increased by the difference between the clear bar distance and 4 ϕ.

c	concrete cover of post-installed fischer rebar anchor FRA
c_1	concrete cover at end-face of existing rebar
c_{min}	minimum concrete cover according to **Table B5.1** and to EN 1992-1-1:2011, Section 4.4.1.2
ϕ	nominal diameter of reinforcing bar
l_0	lap length, according to EN 1992-1-1:2011, Section 8.7.3
$l_{e,ges}$	overall embedment depth, $\geq l_0 + l_e$
d_0	nominal drill bit diameter, see **Annex B 6**
l_e	length of the bonded in threaded part
t_{fix}	thickness of the fixture
l_v	effective embedment depth

Figures not to scale

Rebar connection with injection system FIS V Plus	**Annex B 4**
Intended use General construction rules for post-installed fischer rebar anchors FRA	

Table B5.1: **Minimum concrete cover** c_{min}[1)] depending of the drilling method and the drilling tolerance

Drilling method	nominal diameter of reinforcing bar ϕ [mm]	Minimum concrete cover c_{min} Without drilling aid [mm]	Minimum concrete cover c_{min} With drilling aid [mm]	
Hammer drilling with standard drill bit or hollow drill bit	< 25	30 mm + 0,06 $l_v \geq 2\ \phi$	30 mm + 0,02 $l_v \geq 2\ \phi$	 Drilling aid
	≥ 25	40 mm + 0,06 $l_v \geq 2\ \phi$	40 mm + 0,02 $l_v \geq 2\ \phi$	
Compressed air drilling	< 25	50 mm + 0,08 l_v	50 mm + 0,02 l_v	
	≥ 25	60 mm + 0,08 $l_v \geq 2\ \phi$	60 mm + 0,02 $l_v \geq 2\ \phi$	

[1)] See Annex B 3, figure B3.1 and Annex B 4, figure B4.1
Note: The minimum concrete cover as specified in EN 1992-1-1:2011 must be observed.

Table B5.2: Dispensers and cartridge sizes corresponding to maximum embedment depth $l_{v,max}$

reinforcing bars (rebar)	fischer rebar anchor FRA	Manual dispenser	Accu and pneumatic dispenser (small)	Accu and pneumatic dispenser (large)
		Cartridge size < 500 ml	Cartridge size < 500 ml	Cartridge size > 500 ml
ϕ [mm]	thread [-]	$l_{v,max}$ / $l_{e,ges,max}$ [mm]	$l_{v,max}$ / $l_{e,ges,max}$ [mm]	$l_{v,max}$ / $l_{e,ges,max}$ [mm]
8	---	1000	1000	1800
10	---			
12	FRA M12 FRA HCR M12		1200	
14	---			
16	FRA M16 FRA HCR M16		1500	
20	FRA M20 FRA HCR M20	700	1300	
25	FRA M24 FRA HCR M24		1000	2000
28	---	700	700	

Table B5.3: Conditions for use **static mixer** without an **extension tube**

Nominal drill hole diameter	d_0	[mm]	10	12	14	16	18	20	25	30	35
Drill hole depth h_0 by using	FIS MR Plus	[mm]	≤ 90		≤ 120	≤ 140	≤ 150	≤ 160	≤ 210		
	FIS JMR	[mm]	-	-	≤ 90	≤ 160	≤ 180	≤ 190	≤ 220	≤ 250	

Rebar connection with injection system FIS V Plus

Annex B 5

Intended use
Minimum concrete cover;
dispenser and cartridge sizes corresponding to maximum embedment depth

Deutsches Institut für Bautechnik DIBt

Table B6.1: Working times t_{work} and curing times t_{cure}

Temperature in the anchorage base [°C]	Maximum working time[1] t_{work}		Minimum curing time[2] t_{cure}	
	FIS V Plus	**FIS VS Plus Low Speed**	**FIS V Plus**	**FIS VS Plus Low Speed**
0 to 5 [3]	13 min	---	3 h	6 h
> 5 to 10 [3]	9 min	20 min	90 min	3 h
> 10 to 20	5 min	10 min	60 min	2 h
> 20 to 30	4 min	6 min	45 min	60 min
> 30 to 40 [4]	2 min	4 min	35 min	60 min

[1] Maximum time from the beginning of the injection to rebar / fischer rebar anchor FRA setting and positioning.
[2] For wet concrete the curing time must be doubled.
[3] If the temperature in the concrete falls below 10 °C the cartridge must be warmed up to +15 °C.
[4] If the temperature in the concrete exceeds 30 °C the cartridge must be cooled down to +15 °C up to 20 °C.

Table B6.2: Installation tools for drilling and cleaning the bore hole and injection of the mortar

reinforcing bars (rebar)	fischer rebar anchor FRA	Drilling and cleaning				Injection	
		Nominal drill bit diameter	Diameter of cutting edge	Steel brush diameter	Diameter of fischer compressed-air nozzle	Diameter of extension tube	Injection adapter
ϕ [mm]	Designation	d_0 [mm]	d_{cut} [mm]	d_b [mm]	[mm]	[mm]	[colour]
8[1]	---	10	≤ 10,50	11,0	---	9	---
		12	≤ 12,50	12,5	11		nature
10[1]	---	12	≤ 12,50	12,5			
		14	≤ 14,50	15			blue
12[1]	FRA M12[1] FRA HCR M12[1]	14	≤ 14,50	15	15		
		16	≤ 16,50	17			red
14	---	18	≤ 18,50	19		9 or 15	yellow
16	FRA M16 FRA HCR M16	20	≤ 20,55	21,5	19		green
20	FRA M20 FRA HCR M20	25	≤ 25,55	26,5			black
25[1]	FRA M24[1] FRA HCR M24[1]	30	≤ 30,55	32	28		grey
		35	≤ 35,70	37			brown
28	---	35	≤ 35,70	37			brown

[1] Both drill bit diameters can be used.

Rebar connection with injection system FIS V Plus

Intended use
Working times and curing times;
Installation tools for drilling and cleaning the bore hole and injection of the mortar

Annex B 6

Safety regulations

Review the Safety Data Sheet (SDS) before use for proper and safe handling!

Wear well-fitting protective goggles and protective gloves when working with mortar FIS V Plus / FIS VS Plus Low Speed.

Important: Observe the instructions for use provided with each cartridge.

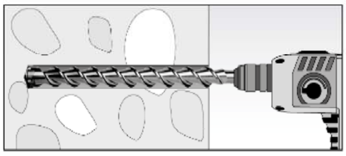
Installation instruction part 1; Installation with FIS V Plus / FIS VS Plus Low Speed

Hole drilling

Note: Before drilling, remove carbonized concrete; clean contact areas (see **Annex B 2**)
In case of aborted drill holes the drill hole shall be filled with mortar.

1a

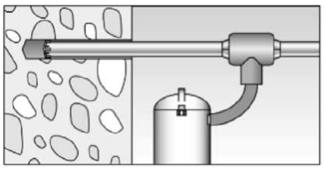
Hammer drilling or compressed air drilling



Drill the hole to the required embedment depth using a hammer drill with carbide drill bit set in rotation hammer mode or a pneumatic drill.
Drill bit sizes see **Table B6.2**.

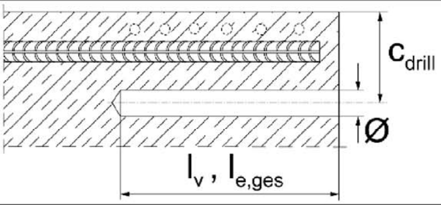
1b

Hammer drilling with hollow drill bit

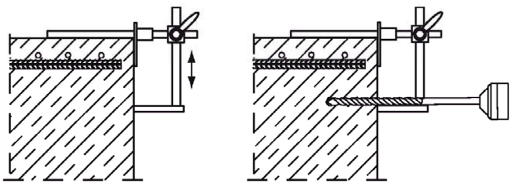


Drill the hole to the required embedment depth using a hammer drill with hollow drill bit in rotation hammer mode.
Dust extraction conditions see drill hole cleaning **Annex B 8**.
Drill bit sizes see **Table B6.2**.

2



Measure and control concrete cover c
(c_{drill} = c + Ø / 2)
Drill parallel to surface edge and to existing rebar.
Where applicable use drilling aid.



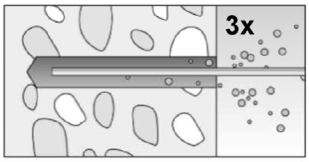
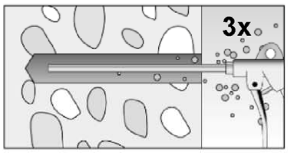
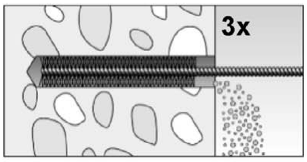
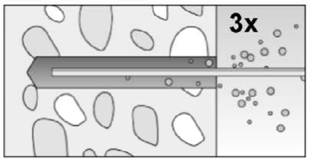
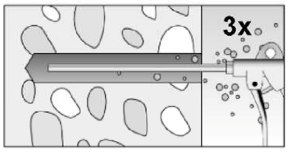
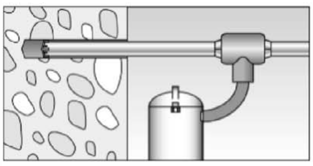
For holes l_v > 20 cm use drilling aid.
Three different options can be considered:

A) drilling aid
B) Slat or spirit level
C) Visual check

Minimum concrete cover c_{min} see **Table B5.1.**

Installation instruction part 2; Installation with FIS V Plus / FIS VS Plus Low Speed

Drill hole cleaning

3a	**Hammer or compressed air drilling** 	
	 Clean the drill hole: For d_0 < 18 mm and depths l_v resp. $l_{e,ges} \le 12 \cdot \phi$ blow out the hole three times by hand.	 For d_0 > 18 mm and depths l_v resp. $l_{e,ges} > 12 \cdot \phi$ blow out the hole three times with oil-free compressed air ($p \ge 6$ bar). Use suitable compressed-air nozzle (see **Table B6.2**).
	 Brush drill hole three times; for drill hole diameters $d_0 \ge 30$ mm attach brush to a power tool and brush hole with a speed of max. 550 revolutions per minute. For deep holes a brush extension is mandatory. Use suitable brushes (see **Table B6.2**).	
	 Clean the drill hole: For d_0 < 18 mm and depths l_v resp. $l_{e,ges} \le 12 \cdot \phi$ blow out the hole three times by hand.	 For d_0 > 18 mm and depths l_v resp. $l_{e,ges} > 12 \cdot \phi$ blow out the hole three times with oil-free compressed air ($p \ge 6$ bar) Use suitable compressed-air nozzle (see **Table B6.2**).
3b	**Hammer drilling with hollow drill bit** 	
	 Use a suitable dust extraction system, e. g. fischer FVC 35 M or a comparable dust extraction system with equivalent performance data. Drill the hole with hollow drill bit. The dust extraction system has to extract the drill dust nonstop during the drilling process and must be adjusted to maximum power. No further drill hole cleaning necessary.	

Rebar connection with injection system FIS V Plus	**Annex B 8**
Intended use Installation instruction part 2, drill hole cleaning	

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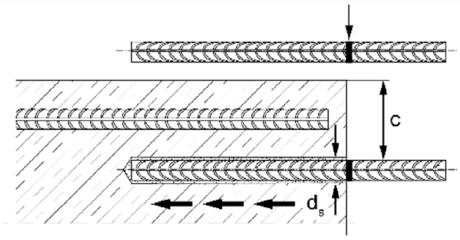
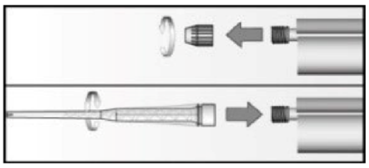
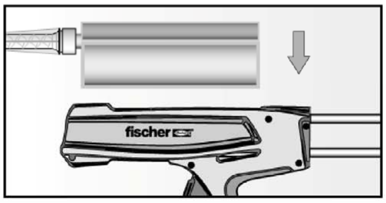
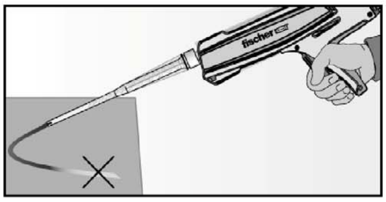
Installation instruction part 3; Installation with FIS V Plus / FIS VS Plus Low Speed
reinforcing bars (rebar) / fischer rebar anchor FRA and cartridge preparation

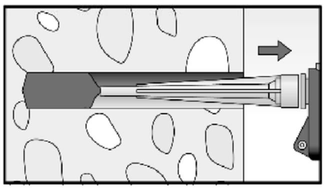
4		Before use, make asure that the rebar or the fischer rebar anchor FRA is dry and free of oil or other residue. Mark the embedment depth l_v (e.g. with tape) Insert rebar in borehole, to verify drill hole depth and setting depth l_v resp. $l_{e,ges.}$
5		Twist off the sealing cap Twist on the static mixer (the spiral in the static mixer must be clearly visible).
6		Place the cartridge into a suitable dispenser.
7		Press out approximately 10 cm of mortar until the resin is permanently grey in colour. Mortar which is not grey in colour will not cure and must be disposed.

Rebar connection with injection system FIS V Plus	**Annex B 9**
Intended use Installation instruction part 3, reinforcing bars (rebar) / fischer rebar anchor FRA and cartridge preparation	

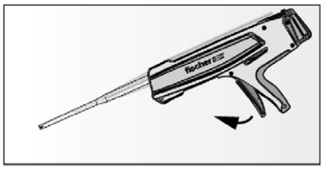
Installation instruction part 4; Installation with FIS V Plus / FIS VS Plus Low Speed

Injection of the mortar without extension tube

8a



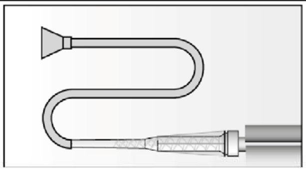
Inject the mortar from the back of the hole towards the front and slowly withdraw the static mixer step by step with each trigger pull.
Avoid bubbles.
Fill holes approximately 2/3 full, to ensure that the annular gap between the rebar and the concrete will be completely filled with adhesive over the entire embedment length.
The conditions for mortar injection without extension tube can be found in **Table B5.3**.



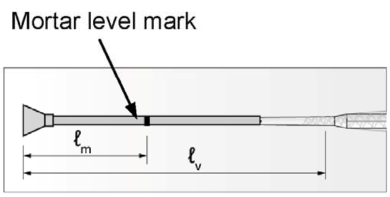
After injecting, release the dispenser. This will prevent further mortar discharge from the static mixer.

Injection of the mortar with extension tube

8b



Assemble mixing nozzle FIS MR Plus or FIS JMR, extension tube and appropriate injection adapter (see **Table B6.2**).



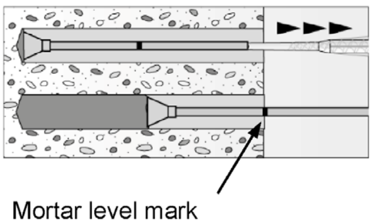
Mark the required mortar level l_m and embedment depth l_v resp. $l_{e,ges}$ with tape or marker on the injection extension tube.

a) Estimation:

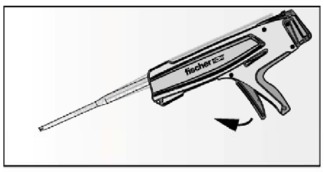
$$l_m = \frac{1}{3} \cdot l_v \; resp. \; l_m = \frac{1}{3} \cdot l_{e,ges} \text{ [mm]}$$

b) Precise equation for optimum mortar volume:

$$l_m = l_v \; resp. \; l_{e,ges} \left(\left(1{,}2 \cdot \frac{d_s^2}{d_0^2} - 0{,}2\right) \right) \text{ [mm]}$$



Insert injection adapter to back of the hole. Begin injection allowing the pressure of the injected adhesive mortar to push the injection adapter towards the front of the hole. Do not actively pull out!
Fill holes approximately 2/3 full, to ensure that the annular gap between the rebar and the concrete will be completely filled with adhesive over the embedment length.
When using an injection adapter continue injection until the mortar level mark l_m becomes visible.
Maximum embedment depth see **Table B5.2**.

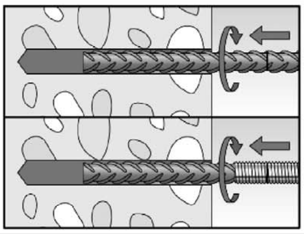
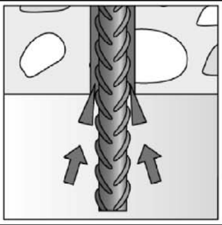
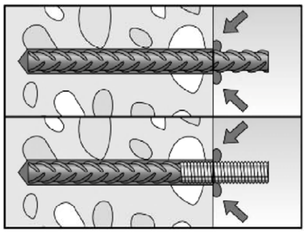
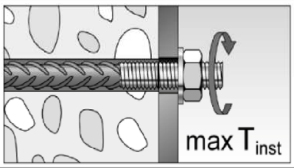


After injecting, release the dispenser. This will prevent further mortar discharge from static mixer.

Rebar connection with injection system FIS V Plus	**Annex B 10**
Intended use Installation instruction part 4, mortar injection	

Installation instruction part 5; Installation with FIS V Plus / FIS VS Plus Low Speed

Insert rebar / fischer rebar anchor FRA

9		Insert the rebar / fischer rebar anchor FRA slowly twisted into the borehole until the embedment mark is reached. Recommendation: Rotation back and forth of the reinforcement bar or the fischer rebar anchor FRA makes pushing easy.
10		For overhead installation, support the rebar / fischer rebar anchor FRA and secure it from falling till mortar started to harden, e.g. using wedges.
11		After installing the rebar or fischer rebar anchor FRA the annular gap must be completely filled with mortar. Proper installation • Desired embedment depth is reached l_v, resp. $l_{e,ges}$: embedment mark at concrete surface • Excess mortar flows out of the borehole after the rebar has been fully inserted up to the embedment mark.
12		Observe the working time "t_{work}" (**see Table B6.1**), which varies according to temperature of base material. Minor adjustments to the rebar / fischer rebar anchor FRA position may be performed during the working time Full load may be applied only after the curing time "t_{cure}" has elapsed (**see Table B 6.1**).
13		Mounting the fixture for fischer rebar anchor FRA, max T_{inst} see T**able A6.1**.

Rebar connection with injection system FIS V Plus	**Annex B 11**
Intended use Installation instruction part 5, insert rebar / fischer rebar anchor FRA	

Minimum anchorage length and minimum lap length

The minimum anchorage length $l_{b,min}$ and the minimum lap length $l_{0,min}$ according to EN 1992-1-1:2011 shall be multiplied by the relevant amplification factor α_{lb} according to **Table C1.1.**

Table C1.1: Amplification factor α_{lb} related to concrete strength class and drilling method

Hammer drilling, hollow drilling and compressed air drilling

Rebar / fischer rebar anchor FRA	Amplification factor α_{lb}								
	Concrete strength class								
ϕ [mm]	C12/15	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8 to 25	1,0						1,1		1,2
28	1,0								

Table C1.2: Bond efficiency factor k_b related to concrete strength class and drilling method

Hammer drilling, hollow drilling and compressed air drilling

Rebar / fischer rebar anchor FRA	Bond efficiency factor k_b								
	Concrete strength class								
ϕ [mm]	C12/15	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8 to 25	1,00								
28	1,00						0,91	0,84	0,84

Table C1.3: Design values of the bond strength $f_{bd,PIR}$ in N/mm² related to concrete strength class and drilling method for good bond conditions

$$f_{bd,PIR} = k_b \cdot f_{bd}$$

f_{bd}: Design value of the bond strength in N/mm² considering the concrete strength classes and the rebar diameter for good bond condition (for all other bond conditions multiply the values by $\eta_1 = 0{,}7$)
and recommended partial factor $\gamma_c = 1{,}5$ according to EN 1992-1-1:2011

k_b: Bond efficiency factor according to **Table C1.2**

Hammer drilling, hollow drilling and compressed air drilling

Rebar / fischer rebar anchor FRA	Bond strength $f_{bd,PIR}$ [N/mm²]								
	Concrete strength class								
ϕ [mm]	C12/15	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8 to 25	1,6	2,0	2,3	2,7	3,0	3,4	3,7	4,0	4,3
28	1,6	2,0	2,3	2,7	3,0	3,4	3,4	3,4	3,7

Rebar connection with injection system FIS V Plus	Annex C 1
Performance Amplification factor α_{lb}, bond efficiency factor k_b, design values of the bond strength $f_{bd,PIR}$	

Table C2.1: Characteristic **tensile yield strength** for rebar part of **fischer rebar anchors FRA**

fischer rebar anchor FRA / FRA HCR			**M12**	**M16**	**M20**	**M24**
Characteristic tensile yield strength for rebar part						
Rebar diameter	ϕ	[mm]	12	16	20	25
Characteristic tensile yield strength for rebar	f_{yk}	[N/mm²]	500	500	500	500
Partial factor for rebar part	$\gamma_{Ms,N}$ 1)	[-]	1,15			

1) In absence of national regulations

Table C2.2: Characteristic resistance to **steel failure** under tension loading of **fischer rebar anchors FRA**

fischer rebar anchor FRA / FRA HCR			**M12**	**M16**	**M20**	**M24**
Characteristic resistance to steel failure under tension loading						
Characteristic resistance	$N_{Rk,s}$	[kN]	62	111	173	263
Partial factor						
Partial factor	$\gamma_{Ms,N}$ 1)	[-]	1,4			

1) In absence of national regulations

Table C2.3: Characteristics resistance to **steel failure** for **fischer rebar anchors FRA** under tension loading and fire exposure R30 to R120

fischer rebar anchor FRA / FRA HCR				**M12**	**M16**	**M20**	**M24**
Characteristic resistance to steel failure under tension loading and fire exposure	R30	$N_{Rk,s,fi}$	[kN]	2,5	4,7	7,4	10,6
	R60			2,1	3,9	6,1	8,8
	R90			1,7	3,1	4,9	7,1
	R120			1,3	2,5	3,9	5,6

Design value of the ultimate bond strength $f_{bd,fi}$ at increased temperature for concrete strength classes C12/15 to C50/60 (all drilling methods)

The design value of the bond strength $f_{bd,fi}$ at increased temperature has to be calculated by the following equation:

$$f_{bd,fi} = k_{fi}(\theta) \cdot f_{bd,PIR} \cdot \frac{\gamma_c}{\gamma_{m,fi}}$$

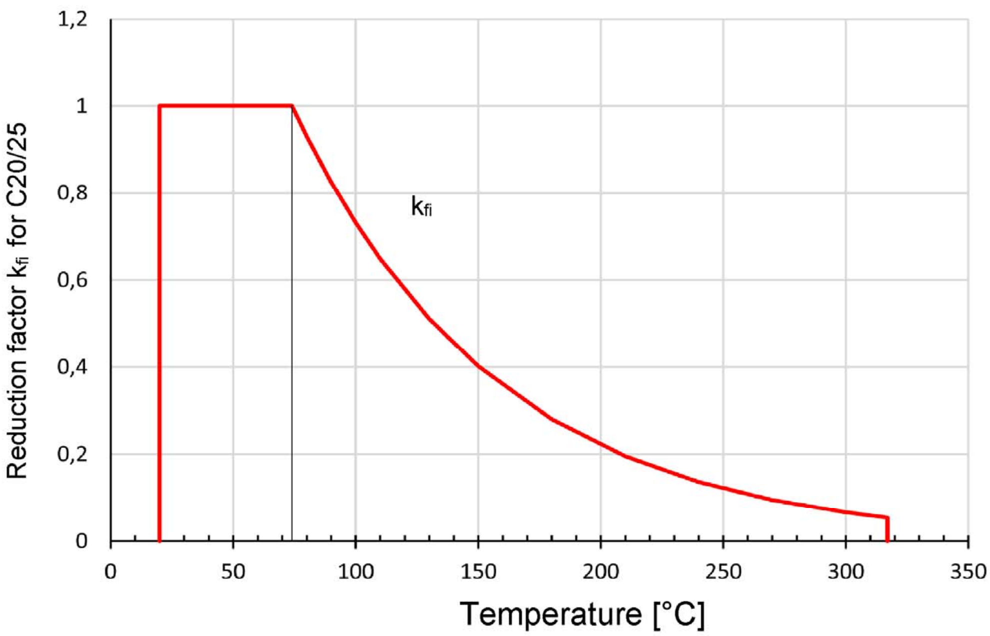
If: $\theta > 74$ °C $\quad k_{fi}(\theta) = \frac{24{,}308 \cdot e^{-0{,}012 \cdot \theta}}{f_{bd,PIR} \cdot 4{,}3} \leq 1{,}0$

If: $\theta > \theta_{max}$ (317 °C) $\quad k_{fi}(\theta) = 0$

$f_{bd,fi}$	=	Design value of the ultimate bond strength at increased temperature in N/mm²
θ	=	Temperature in °C in the mortar layer
$k_{fi}(\theta)$	=	Reduction factor at increased temperature
$f_{bd,PIR}$	=	Design value of the bond strength in N/mm² in cold condition according to **Table C1.3** considering the concrete strength classes, the rebar diameter, the drilling method and the bond conditions according to EN 1992-1-1:2011
γ_C	=	1,5 recommended partial factor according to EN 1992-1-1:2011
$\gamma_{m,fi}$	=	1,0 recommended partial factor

For evidence at increased temperature the anchorage length shall be calculated according to EN 1992-1-1:2011 Equation 8.3 using the temperature-dependent ultimate design value of bond strength $f_{bd,fi}$.

Figure C3.1: Example graph of reduction factor $k_{fi}(\theta)$ for concrete class C20/25 for good bond conditions



Rebar connection with injection system FIS V Plus	**Annex C 3**
Performance Design value of bond strength $f_{bd,fi}$ at increased temperature	

Deutsches Institut für Bautechnik

DIBt

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Approval body for construction products and types of construction

Bautechnisches Prüfamt

An institution established by the Federal and Laender Governments

Designated according to Article 29 of Regulation (EU) No 305/2011 and member of EOTA (European Organisation for Technical Assessment)

European Technical Assessment — ETA-20/0603 of 13 November 2020

English translation prepared by DIBt - Original version in German language

General Part

Technical Assessment Body issuing the European Technical Assessment:	Deutsches Institut für Bautechnik
Trade name of the construction product	fischer Injection system FIS V Plus
Product family to which the construction product belongs	Bonded anchor for use in concrete
Manufacturer	fischerwerke GmbH & Co. KG Otto-Hahn-Straße 15 79211 Denzlingen DEUTSCHLAND
Manufacturing plant	fischerwerke
This European Technical Assessment contains	37 pages including 3 annexes which form an integral part of this assessment
This European Technical Assessment is issued in accordance with Regulation (EU) No 305/2011, on the basis of	EAD 330499-01-0601 Edition 04/2020

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Specific Part

1 Technical description of the product

The "fischer Injection system FIS V Plus" is a bonded anchor consisting of a cartridge with injection mortar according to Annex A 4 and a steel element according to Annex A 1 to A 3.

The steel element is placed into a drilled hole filled with injection mortar and is anchored via the bond between metal part, injection mortar and concrete.

The product description is given in Annex A.

2 Specification of the intended use in accordance with the applicable European Assessment Document

The performances given in Section 3 are only valid if the anchor is used in compliance with the specifications and conditions given in Annex B.

The verifications and assessment methods on which this European Technical Assessment is based lead to the assumption of a working life of the anchor of at least 50 and/or 100 years. The indications given on the working life cannot be interpreted as a guarantee given by the producer, but are to be regarded only as a means for choosing the right products in relation to the expected economically reasonable working life of the works.

3 Performance of the product and references to the methods used for its assessment

3.1 Mechanical resistance and stability (BWR 1)

Essential characteristic	Performance
Characteristic resistance to tension load (static and quasi-static loading)	See Annex C 1, C 2, C 4 to C 9, B 4, B 5
Characteristic resistance to shear load (static and quasi-static loading)	See Annex C 1 to C 3
Displacements under short-term and long-term loading	See Annex C 10 to C 11
Characteristic resistance and displacements for seismic performance categories C1 and C2	See Annex C 12 to C 15

3.2 Hygiene, health and the environment (BWR 3)

Essential characteristic	Performance
Content, emission and/or release of dangerous substances	No performance assessed

4 Assessment and verification of constancy of performance (AVCP) system applied, with reference to its legal base

In accordance with the European Assessment Document EAD 330499-01-0601 the applicable European legal act is: [96/582/EC].

The system to be applied is: 1

5 Technical details necessary for the implementation of the AVCP system, as provided for in the applicable European Assessment Document

Technical details necessary for the implementation of the AVCP system are laid down in the control plan deposited at Deutsches Institut für Bautechnik.

Issued in Berlin 13 November 2020 by Deutsches Institut für Bautechnik

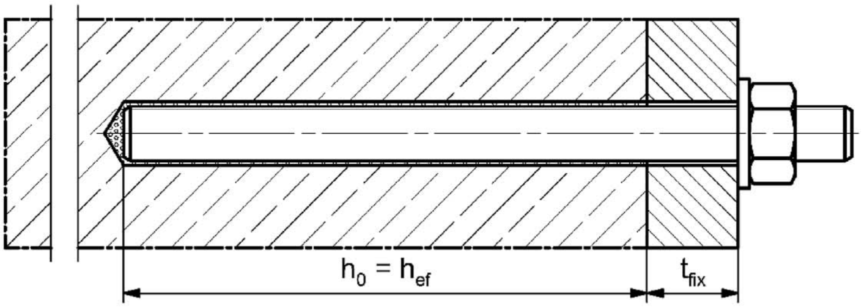
Dipl.-Ing. Beatrix Wittstock
Head of Section

beglaubigt:
Lange

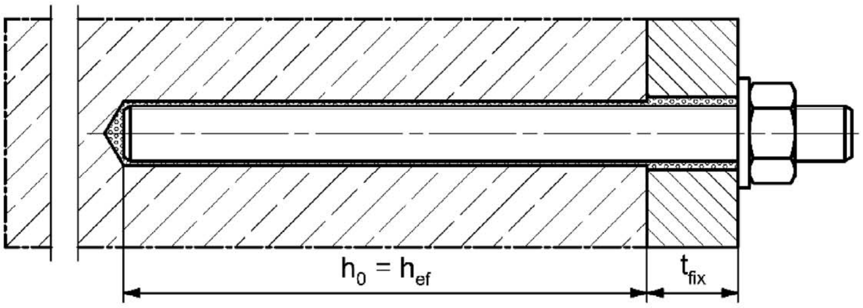
Installation conditions part 1

fischer anchor rod

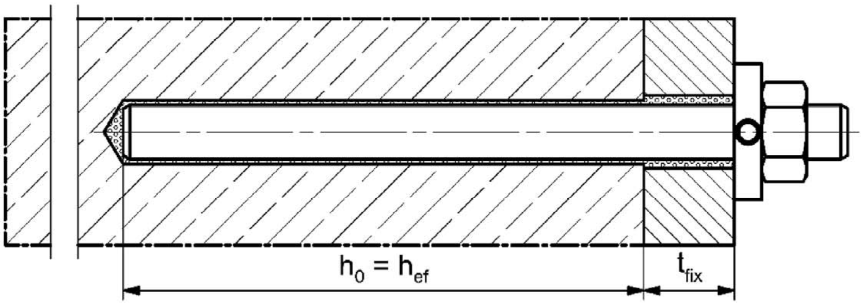
Pre-positioned installation



Push through installation (annular gap filled with mortar)



Pre-positioned or push through installation with subsequently injected fischer filling disc
(annular gap filled with mortar)



Figures not to scale

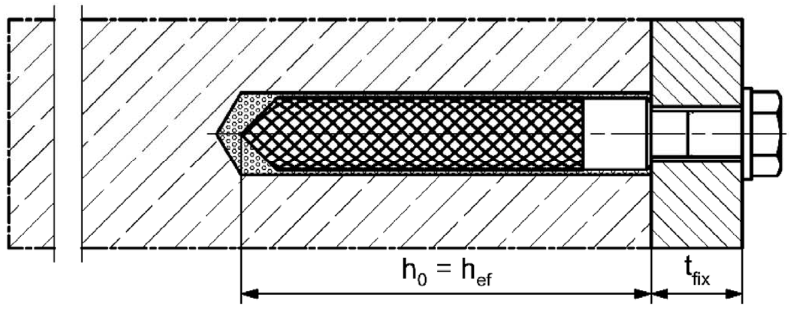
h_0 = drill hole depth
h_{ef} = effective embedment depth
t_{fix} = thickness of fixture

fischer injection system FIS V Plus	Annex A 1
Product description Installation conditions part 1	

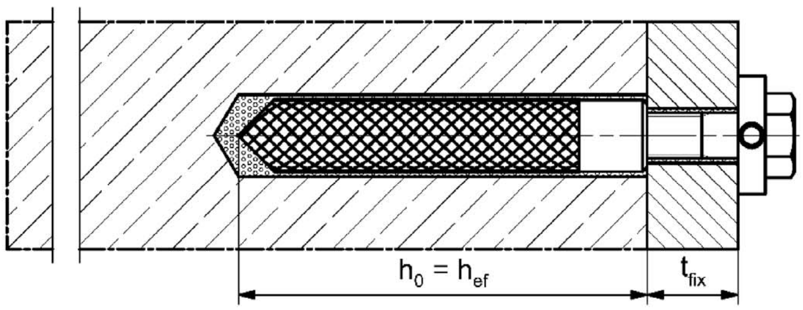
Installation conditions part 2

fischer internal threaded anchor RG MI

Pre-positioned installation



Pre-positioned installation with subsequently injected fischer filling disc (annular gap filled with mortar)



Figures not to scale

h_0 = drill hole depth

t_{fix} = thickness of fixture

h_{ef} = effective embedment depth

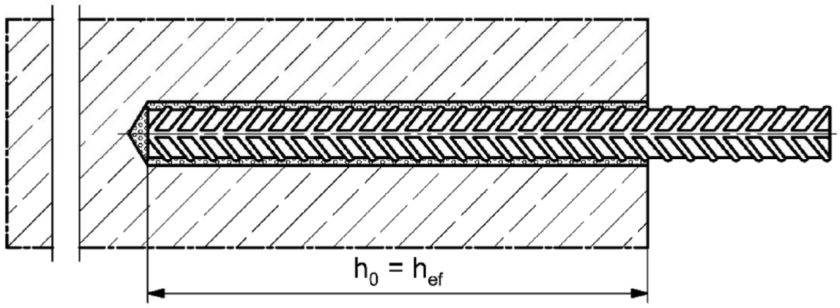
fischer injection system FIS V Plus

Product description
Installation conditions part 2

Annex A 2

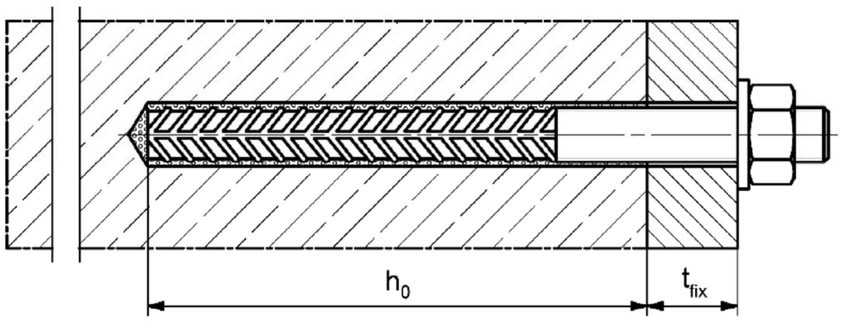
Installation conditions part 3

Reinforcing bar

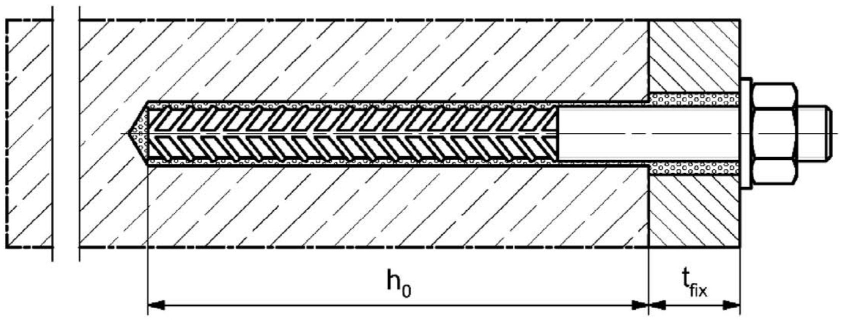


fischer rebar anchor FRA

Pre-positioned installation



Push through installation (annular gap filled with mortar)



Figures not to scale

h_0 = drill hole depth

h_{ef} = effective embedment depth

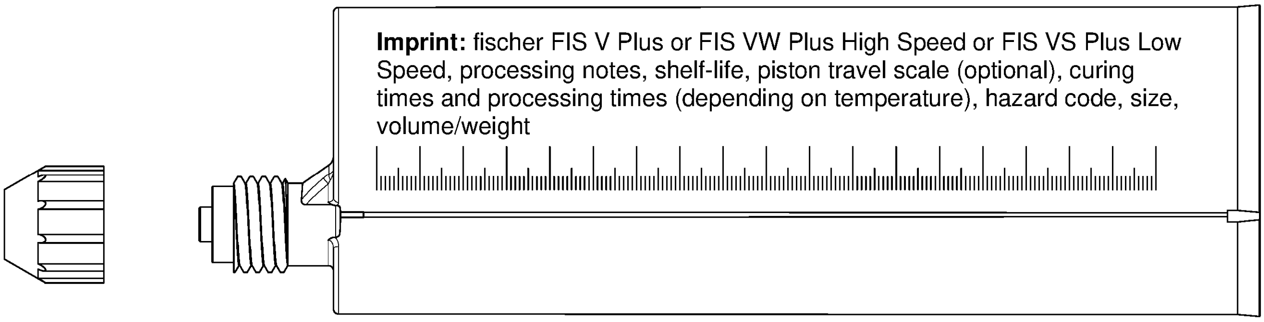
t_{fix} = thickness of fixture

fischer injection system FIS V Plus	**Annex A 3**
Product description Installation conditions part 3	

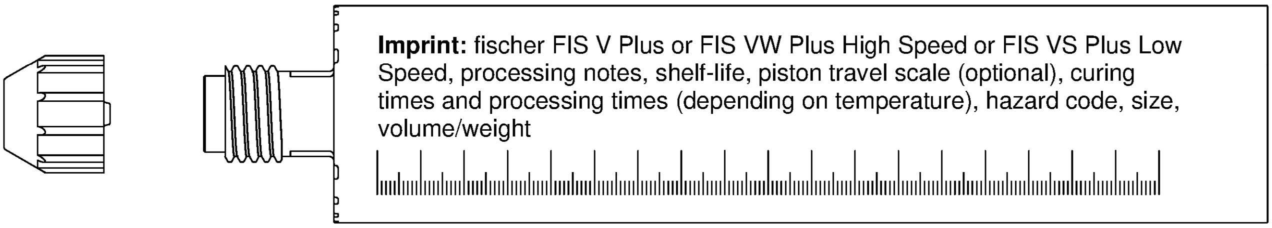
Deutsches Institut für Bautechnik DIBt

Overview system components part 1

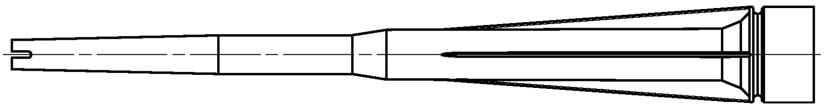
Injection cartridge (shuttle cartridge) with sealing cap; Sizes: 350 ml, 360 ml, 390 ml, 550 ml, 1100 ml, 1500 ml



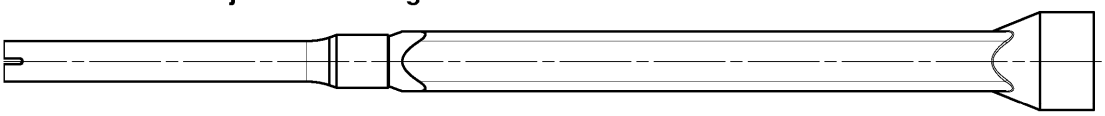
Injection cartridge (coaxial cartridge) with sealing cap; Sizes: 100 ml, 150 ml, 300 ml, 380 ml, 400 ml, 410 ml



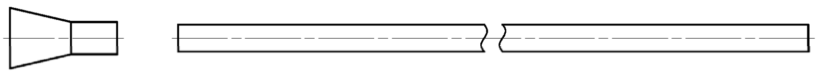
Static mixer FIS MR Plus for injection cartridges up to 410 ml



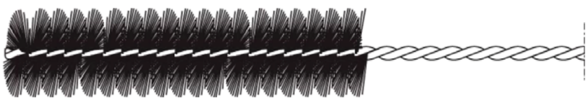
Static mixer FIS UMR for injection cartridges from 550 ml



**Injection adapter and extension tube Ø 9 for static mixer FIS MR Plus;
Injection adapter and extension tube Ø 9 or Ø 15 for static mixer FIS UMR**



Cleaning brush BS



Blow-out pump

AB G:

ABP:

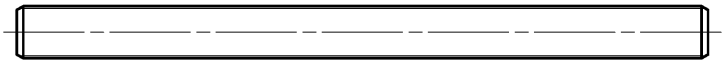
Figures not to scale

fischer injection system FIS V Plus	**Annex A 4**
Product description Overview system components part 1; cartridges / static mixer / accessories	

Overview system components part 2

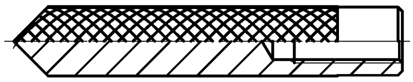
fischer anchor rod

Size: M6, M8, M10, M12, M16, M20, M24, M27, M30

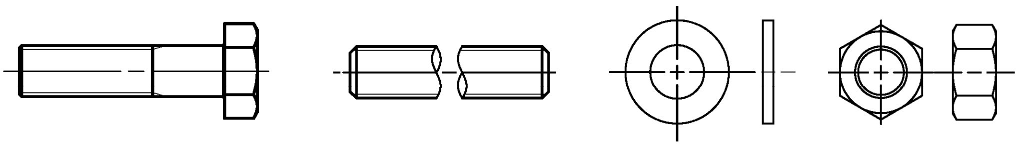


fischer internal threaded anchor RG MI

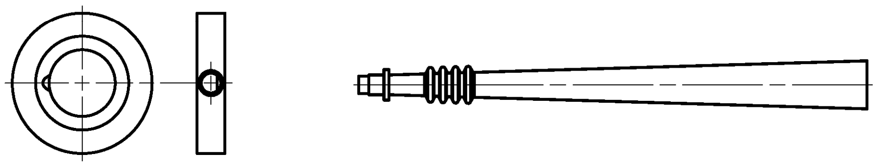
Size: M8, M10, M12, M16, M20



Screw / threaded rod / washer / hexagon nut

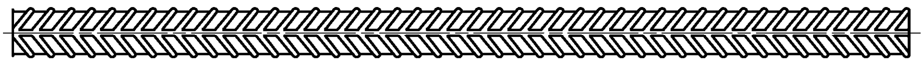


fischer filling disc with injection adapter



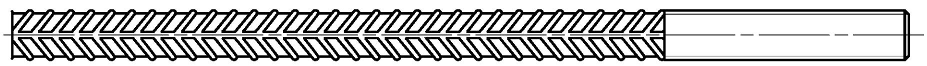
Reinforcing bar

Nominal diameter: ϕ8, ϕ10, ϕ12, ϕ14, ϕ16, ϕ20, ϕ25, ϕ28



fischer rebar anchor FRA

Size: M12, M16, M20, M24



Figures not to scale

fischer injection system FIS V Plus	**Annex A 5**
Product description Overview system components part 2; metal parts, injection adapter	

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Table A6.1: Materials

Part	Designation	Material		
1	Injection cartridge	Mortar, hardener, filler		
	Steel grade	Steel	Stainless steel R	High corrosion resistant steel HCR
		zinc plated	acc. to EN 10088-1:2014 Corrosion resistance class CRC III acc. to EN 1993-1-4:2015	acc. to EN 10088-1:2014 Corrosion resistance class CRC V acc. to EN 1993-1-4:2015
2	Anchor rod	Property class 4.8, 5.8 or 8.8; EN ISO 898-1:2013 zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K) or hot dip galvanised ≥ 40 µm EN ISO 10684:2004 f_{uk} ≤ 1000 N/mm² A_5 > 12% fracture elongation	Property class 50, 70 or 80 EN ISO 3506-1:2009 1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; 1.4062, 1.4662, 1.4462; EN 10088-1:2014 f_{uk} ≤ 1000 N/mm² A_5 > 12% fracture elongation	Property class 50 or 80 EN ISO 3506-1:2009 or property class 70 with f_{yk}= 560 N/mm² 1.4565; 1.4529; EN 10088-1:2014 f_{uk} ≤ 1000 N/mm² A_5 > 12% fracture elongation
		Fracture elongation A_5 > 8 %, for applications without requirements for seismic performance category C2		
3	Washer ISO 7089:2000	zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K) or hot dip galvanised ≥ 40 µm EN ISO 10684:2004	1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; EN 10088-1:2014	1.4565; 1.4529; EN 10088-1:2014
4	Hexagon nut	Property class 4, 5 or 8; EN ISO 898-2:2012 zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K) or hot dip galvanised ≥ 40 µm EN ISO 10684:2004	Property class 50, 70 or 80 EN ISO 3506-1:2009 1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; EN 10088-1:2014	Property class 50, 70 or 80 EN ISO 3506-1:2009 1.4565; 1.4529; EN 10088-1:2014
5	fischer internal threaded anchor RG MI	Property class 5.8 ISO 898-1:2013 zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K)	Property class 70 EN ISO 3506-1:2009 1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; EN 10088-1:2014	Property class 70 EN ISO 3506-1:2009 1.4565; 1.4529; EN 10088-1:2014
6	Commercial standard screw or threaded rod for fischer internal threaded anchor RG MI	Property class 5.8 or 8.8; EN ISO 898-1:2013 zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K) A_5 > 8 % fracture elongation	Property class 70 EN ISO 3506-1:2009 1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; EN 10088-1:2014 A_5 > 8 % fracture elongation	Property class 70 EN ISO 3506-1:2009 1.4565; 1.4529; EN 10088-1:2014 A_5 > 8 % fracture elongation
7	fischer filling disc similar to DIN 6319-G	zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K) or hot dip galvanised ≥ 40 µm EN ISO 10684:2004	1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; EN 10088-1:2014	1.4565;1.4529; EN 10088-1:2014
8	Reinforcing bar EN 1992-1-1:2004 and AC:2010, Annex C	Bars and de-coiled rods, class B or C with f_{yk} and k according to NDP or NCL of according to EN 1992-1-1:2004/NA $f_{uk} = f_{tk} = k \cdot f_{yk}$		
9	fischer rebar anchor FRA	Rebar part: Bars and de-coiled rods class B or C with f_{yk} and k according to NDP or NCL of EN 1992-1-1:2004+AC:2010 $f_{uk} = f_{tk} = k \cdot f_{yk}$	Threaded part: Property class 70 or 80 EN ISO 3506-1:2009 1.4401, 1.4404, 1.4571, 1.4578, 1.4439, 1.4362, 1.4062 acc. to EN 10088-1:2014 Corrosion resistance class CRC III acc. to EN 1993-1-4:2015 1.4565; 1.4529 acc. to EN 10088-1:2014 Corrosion resistance class CRC V acc. to EN 1993-1-4:2015	

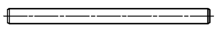
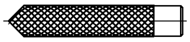
fischer injection system FIS V Plus

Product description
Materials

Annex A 6

Specifications of intended use (part 1)

Table B1.1: Overview use and performance categories

		FIS V Plus with …							
		Anchor rod 		fischer internal threaded anchor RG MI 		Reinforcing bar 		fischer rebar anchor FRA 	
Hammer drilling with standard drill bit		all sizes							
Hammer drilling with hollow drill bit  (fischer „FHD“, Heller „Duster Expert“; Bosch „Speed Clean“; Hilti „TE-CD, TE-YD“, DreBo „D-Plus“, DreBo „D-Max“		Nominal drill bit diameter (d_0) 12 mm to 35 mm							
Static and quasi static load, in	uncracked concrete	all sizes	Tables: C1.1 C4.1 C5.1 C6.1 C10.1	all sizes	Tables: C2.1 C4.1 C7.1 C10.2	all sizes	Tables: C3.1 C4.1 C8.1 C11.1	all sizes	Tables: C3.2 C4.1 C9.1 C11.2
	cracked concrete	M8 to M30		-2)		ϕ 10 to ϕ 28			
Seismic performance category	C1 1)	M10 to M30	Tables: C12.1 C13.1 C14.1	-2)		-2)		-2)	
	C2 1)	M12 M16 M20	Tables: C12.1 C13.1 C15.1						
Use category	I1 dry or wet concrete	all sizes							
	I2 water filled hole	M12 to M30		all sizes		-2)		-2)	
Installation direction		D3 (downward and horizontal and upwards (e.g. overhead) installation)							
Installation temperature		$T_{i,min}$ = -10 °C to $T_{i,max}$ = +40 °C							
In-service temperature	Temperature range I	-40 °C to +80 °C		(max. short term temperature +80 °C; max. long term temperature +50 °C)					
	Temperature range II	-40 °C to +120 °C		(max. short term temperature +120 °C; max. long term temperature +72 °C)					

1) Not for FIS VW Plus High Speed and FIS VS Plus Low Speed
2) No performance assessed

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Specifications of intended use (part 2)

Base materials:

- Compacted reinforced or unreinforced normal weight concrete without fibres of strength classes C20/25 to C50/60 according to EN 206:2013+A1:2016

Use conditions (Environmental conditions):

- Structures subject to dry internal conditions (zinc coated steel, stainless steel or high corrosion resistant steel).
- For all other conditions according to EN1993-1-4:2015 corresponding to corrosion resistance classes to Annex A 6 Table A6.1.

Design:

- Anchorages have to be designed by a responsible engineer with experience of concrete anchor design.
- Verifiable calculation notes and drawings are to be prepared taking account of the loads to be anchored. The position of the anchor is indicated on the design drawings (e. g. position of the anchor relative to reinforcement or to supports, etc.).
- Anchorages are designed in accordance with:
  EN 1992-4:2018 and EOTA Technical Report TR 055, Edition February 2018.

Installation:

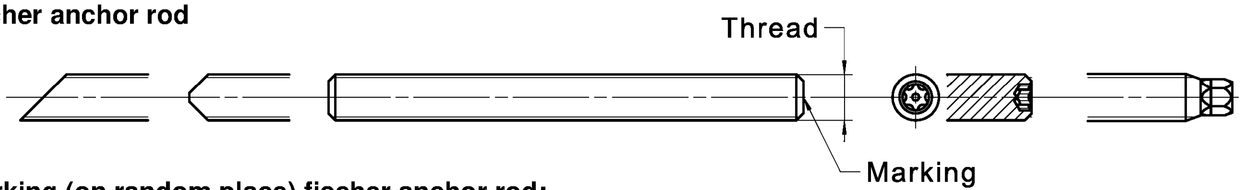
- Anchor installation is to be carried out by appropriately qualified personnel and under the supervision of the person responsible for technical matters of the site
- In case of aborted hole: The hole shall be filled with mortar
- Anchorage depth should be marked and adhered to on installation
- Overhead installation is allowed

Table B3.1: Installation parameters for **anchor rods** [1)]

Anchor rods		**Thread**	**M6**	**M8**	**M10**	**M12**	**M16**	**M20**	**M24**	**M27**	**M30**
Width across flats	SW	[mm]	10	13	17	19	24	30	36	41	46
Nominal drill hole diameter	d_0	[mm]	8	10	12	14	18	24	28	30	35
Drill hole depth	h_0	[mm]	$h_0 = h_{ef}$								
Effective embedment depth	$h_{ef,\,min}$	[mm]	50	60	60	70	80	90	96	108	120
	$h_{ef,\,max}$	[mm]	72	160	200	240	320	400	480	540	600
Minimum spacing and minimum edge distance	$s_{min} = c_{min}$	[mm]	40	40	45	55	65	85	105	125	140
Diameter of the clearance hole of the fixture: pre-positioned installation	d_f	[mm]	7	9	12	14	18	22	26	30	33
Diameter of the clearance hole of the fixture: push through installation	d_f	[mm]	9	12	14	16	20	26	30	33	40
Minimum thickness of concrete member	h_{min}	[mm]	$h_{ef} + 30$ (≥ 100)				$h_{ef} + 2d_0$				
Maximum installation torque	max T_{inst}	[Nm]	5	10	20	40	60	120	150	200	300

[1)] minimum spacing and minimum edge distance see Annex B 4

fischer anchor rod

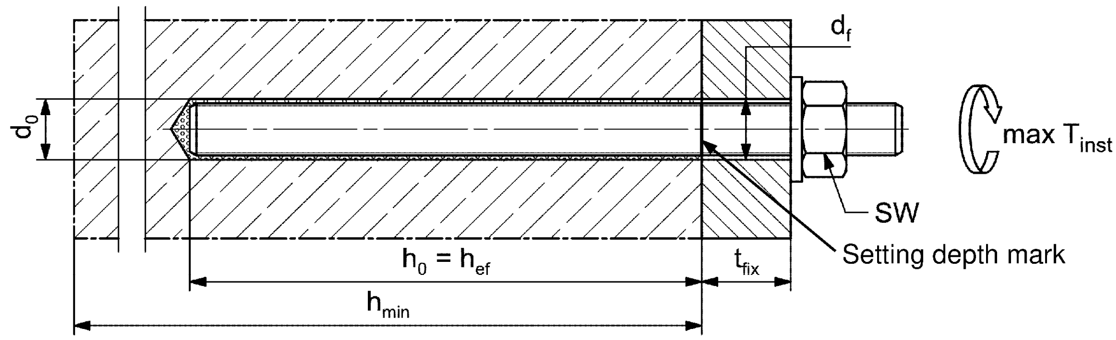


Marking (on random place) fischer anchor rod:

Steel zinc plated PC[1)] 8.8	● or **+**	Steel hot-dip PC[1)] 8.8	●
High corrosion resistant steel HCR PC[1)] 50	●	High corrosion resistant steel HCR PC[1)] 70	**-**
High corrosion resistant steel HCR PC[1)] 80	**(**	Stainless steel R property class 50	~
Stainless steel R property class 80	∗		

Alternatively: Colour coding according to DIN 976-1: 2016 — [1)] PC = property class

Installation conditions:



Commercial standard threaded rods, washers and hexagon nuts may also be used if the following requirements are fulfilled:

- Materials, dimensions and mechanical properties according to Annex A 6, Table A6.1
- Inspection certificate 3.1 according to EN 10204:2004, the documents have to be stored
- Setting depth is marked

Figures not to scale

fischer injection system FIS V Plus	**Annex B 3**
Intended use Installation parameters anchor rods	

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Table B4.1: Minimum spacing and minimum edge distance for **anchor rods**, **reinforcing bars** and **fischer rebar anchor FRA**

Anchor rods			**M6**	**M8**	**M10**	**M12**	**-**	**M16**
Reinforcing bars / FRA (nominal diameter)	ϕ		**-**	**8**	**10**	**12**	**14**	**16**
Minimum edge distance								
Uncracked / cracked concrete	c_{min}	[mm]	40	40	45	45	45	50
Minimum spacing	s_{min}	[mm]	according to Annex B 5					
Minimum spacing								
Uncracked / cracked concrete	s_{min}	[mm]	40	40	45	55	60	65
Minimum edge distance	c_{min}	[mm]	according to Annex B 5					
Required projecting area								
Uncracked concrete	$A_{sp,req}$	[1000 mm²]	8,0	8,0	13,0	22,0	23,0	24,0
Cracked concrete	$A_{sp,req}$	[1000 mm²]	6,5	6,5	10	16,5	17,5	18,5

Anchor rods			**M20**	**M24**	**-**	**M27**	**-**	**M30**
Reinforcing bars / FRA (nominal diameter)	ϕ		**20**	**-**	**25**	**-**	**28**	**-**
Minimum edge distance								
Uncracked / cracked concrete	c_{min}	[mm]	55	60	75	75	80	80
Minimum spacing	s_{min}	[mm]	according to Annex B 5					
Minimum spacing								
Uncracked / cracked concrete	s_{min}	[mm]	85	105	120	120	140	140
Minimum edge distance	c_{min}	[mm]	according to Annex B 5					
Required projecting area								
Uncracked concrete	$A_{sp,req}$	[1000 mm²]	38,5	40	47,5	47,5	64	64
Cracked concrete	$A_{sp,req}$	[1000 mm²]	29,5	30,5	36,5	36,5	49	49

Splitting failure for minimum edge distance and spacing in dependence of the effective embedment depth h_{ef}.

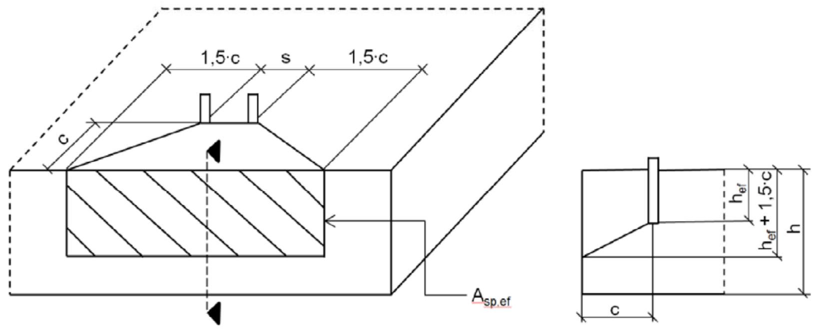
For the calculation of minimum spacing and minimum edge distance of anchors in combination with different embedment depths and thicknesses of concrete members the following equation shall be fulfilled:

$$\mathbf{A_{sp,req} < A_{sp,t}}$$

$A_{sp,req}$ = required projecting area
$A_{sp,t}$ = $A_{sp,ef}$ = effective projecting area (according to Annex B 5)

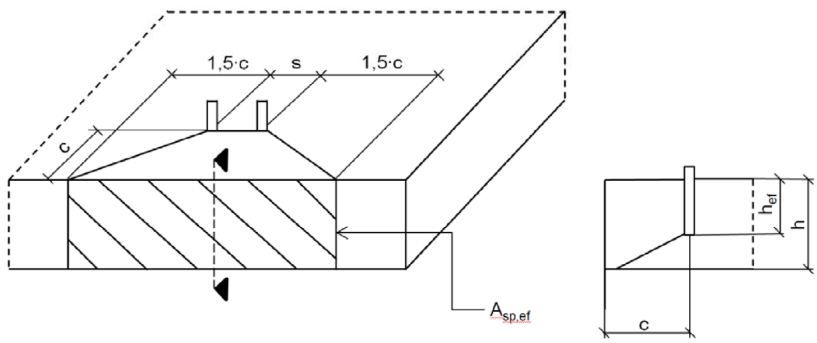
fischer injection system FIS V Plus	**Annex B 4**
Intended use Minimum spacing and edge distance for anchor rods, reinforcing bars and fischer rebar anchor FRA	

Table B5.1: Effective projecting area $A_{sp,t}$ with concrete member thickness $h > h_{ef} + 1{,}5 \cdot c$ and $h \geq h_{min}$



Single anchor		$A_{sp,t} = (3 \cdot c) \cdot (h_{ef} + 1{,}5 \cdot c)$	[mm²]	with $c \geq c_{min}$
Group of anchors with	$s > 3 \cdot c$	$A_{sp,t} = (6 \cdot c) \cdot (h_{ef} + 1{,}5 \cdot c)$	[mm²]	
Group of anchors with	$s \leq 3 \cdot c$	$A_{sp,t} = (3 \cdot c + s) \cdot (h_{ef} + 1{,}5 \cdot c)$	[mm²]	with $c \geq c_{min}$ and $s \geq s_{min}$

Table B5.2: Effektive projecting area $A_{sp,t}$ with concrete member thickness $h \leq h_{ef} + 1{,}5 \cdot c$ and $h \geq h_{min}$



Single anchor		$A_{sp,t} = 3 \cdot c \cdot$ existing h	[mm²]	with $c \geq c_{min}$
Group of anchors with	$s > 3 \cdot c$	$A_{sp,t} = 6 \cdot c \cdot$ existing h	[mm²]	
Group of anchors with	$s \leq 3 \cdot c$	$A_{sp,t} = (3 \cdot c + s) \cdot$ existing h	[mm²]	with $c \geq c_{min}$ and $s \geq s_{min}$

Edge distance and axial spacing shall be rounded up to at least 5 mm

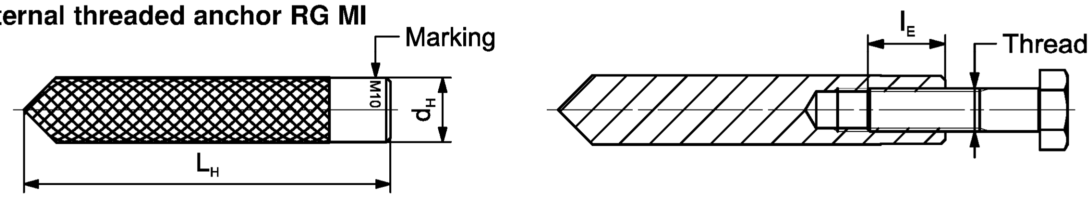
Figures not to scale

fischer injection system FIS V Plus	**Annex B 5**
Intended use Minimum thickness of concrete member for anchor rods and reinforcing bars, minimum spacing and edge distance	

Table B6.1: Installation parameters for **fischer internal threaded anchors RG MI**

Internal threaded anchors RG MI		Thread	M8	M10	M12	M16	M20
Diameter of anchor	$d_{nom} = d_H$	[mm]	12	16	18	22	28
Nominal drill hole diameter	d_0	[mm]	14	18	20	24	32
Drill hole depth	h_0	[mm]	$h_0 = h_{ef} = L_H$				
Effective embedment depth ($h_{ef} = L_H$)	h_{ef}	[mm]	90	90	125	160	200
Minimum spacing and minimum edge distance	$s_{min} = c_{min}$	[mm]	55	65	75	95	125
Diameter of clearance hole in the fixture	d_f	[mm]	9	12	14	18	22
Minimum thickness of concrete member	h_{min}	[mm]	120	125	165	205	260
Maximum screw-in depth	$l_{E,max}$	[mm]	18	23	26	35	45
Minimum screw-in depth	$l_{E,min}$	[mm]	8	10	12	16	20
Maximum installation torque	max T_{inst}	[Nm]	10	20	40	80	120

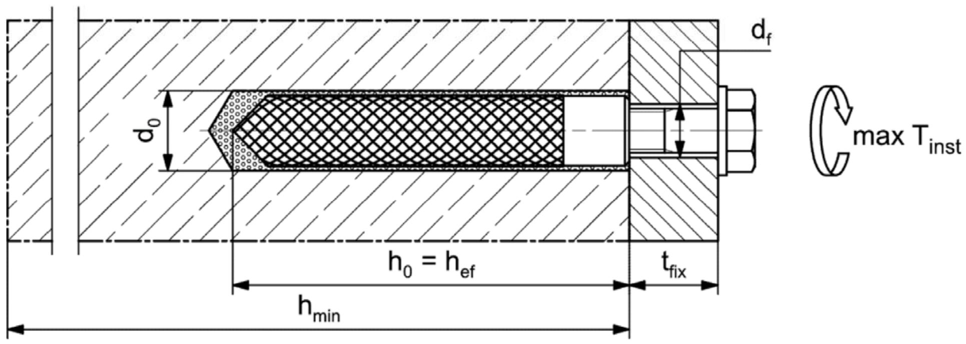
fischer internal threaded anchor RG MI



Marking: Anchor size e. g.: **M10**
Stainless steel → additional **R**; e.g.: **M10 R**
High corrosion resistant steel → additional **HCR**; e.g.: **M10 HCR**

Retaining bolt or threaded rods (including nut and washer) must comply with the appropriate material and strength class of Annex A 6, Table A6.1

Installation conditions:



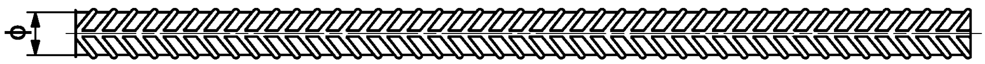
Figures not to scale

fischer injection system FIS V Plus	**Annex B 6**
Intended use Installation parameters internal threaded anchors RG MI	

<table>
<tr><th colspan="14">Table B7.1: Installation parameters for **reinforcing bars** 1)</th></tr>
<tr><th colspan="2">Nominal diameter of the bar</th><th>ϕ</th><th colspan="2">8 2)</th><th colspan="2">10 2)</th><th colspan="2">12 2)</th><th>14</th><th>16</th><th>20</th><th>25</th><th>28</th></tr>
<tr><td>Nominal drill hole diameter</td><td>d_0</td><td rowspan="6">[mm]</td><td>10</td><td>12</td><td>12</td><td>14</td><td>14</td><td>16</td><td>18</td><td>20</td><td>25</td><td>30</td><td>35</td></tr>
<tr><td>Drill hole depth</td><td>h_0</td><td colspan="11">$h_0 = h_{ef}$</td></tr>
<tr><td rowspan="2">Effective embedment depth</td><td>$h_{ef,min}$</td><td colspan="2">60</td><td colspan="2">60</td><td colspan="2">70</td><td>75</td><td>80</td><td>90</td><td>100</td><td>112</td></tr>
<tr><td>$h_{ef,max}$</td><td colspan="2">160</td><td colspan="2">200</td><td colspan="2">240</td><td>280</td><td>320</td><td>400</td><td>500</td><td>560</td></tr>
<tr><td>Minimum spacing and minimum edge distance</td><td>s_{min} = c_{min}</td><td colspan="2">40</td><td colspan="2">45</td><td colspan="2">55</td><td>60</td><td>65</td><td>85</td><td>110</td><td>130</td></tr>
<tr><td>Minimum thickness of concrete member</td><td>h_{min}</td><td colspan="4">$h_{ef} + 30$
(≥ 100)</td><td colspan="7">$h_{ef} + 2d_0$</td></tr>
</table>

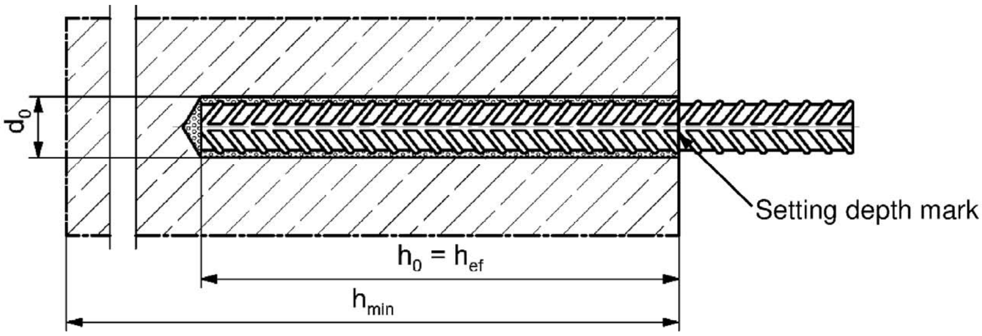
1) minimum spacing and minimum edge distance see Annex B 4
2) Both drill hole diameters can be used

Reinforcing bar



- The minimum value of related rib area $f_{R,min}$ must fulfil the requirements of EN 1992-1-1:2004+AC:2010
- The rib height must be within the range: $0{,}05 \cdot \phi \leq h_{rib} \leq 0{,}07 \cdot \phi$
  (ϕ = Nominal diameter of the bar, h_{rib} = rib height)

Installation conditions:



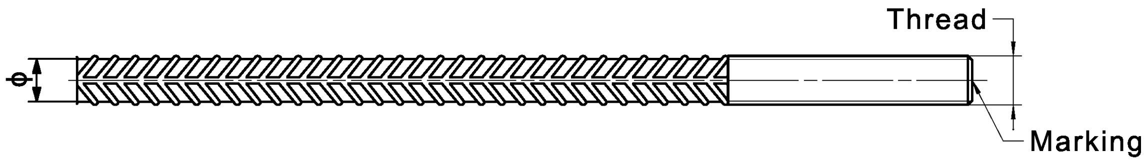
Figures not to scale

fischer injection system FIS V Plus	**Annex B 7**
Intended use Installation parameters reinforcing bars	

<table>
<tr><td colspan="3">Table B8.1: Installation parameters for fischer rebar anchor FRA 1)</td><td colspan="6"></td></tr>
<tr><td colspan="3">Rebar anchor FRA</td><td>Thread</td><td colspan="2">M12 2)</td><td>M16</td><td>M20</td><td>M24</td></tr>
<tr><td colspan="2">Nominal diameter of the bar</td><td>ϕ</td><td rowspan="10">[mm]</td><td colspan="2">12</td><td>16</td><td>20</td><td>25</td></tr>
<tr><td colspan="2">Width across flats</td><td>SW</td><td colspan="2">19</td><td>24</td><td>30</td><td>36</td></tr>
<tr><td colspan="2">Nominal drill hole diameter</td><td>d_0</td><td>14</td><td>16</td><td>20</td><td>25</td><td>30</td></tr>
<tr><td colspan="2">Drill hole depth</td><td>h_0</td><td colspan="5">$h_{ef} + l_e$</td></tr>
<tr><td colspan="2" rowspan="2">Effective embedment depth</td><td>$h_{ef,min}$</td><td colspan="2">70</td><td>80</td><td>90</td><td>96</td></tr>
<tr><td>$h_{ef,max}$</td><td colspan="2">140</td><td>220</td><td>300</td><td>380</td></tr>
<tr><td colspan="2">Distance concrete surface to welded joint</td><td>l_e</td><td colspan="5">100</td></tr>
<tr><td colspan="2">Minimum spacing and minimum edge distance</td><td>$s_{min} = c_{min}$</td><td colspan="2">55</td><td>65</td><td>85</td><td>105</td></tr>
<tr><td rowspan="2">Diameter of clearance hole in the fixture</td><td>pre-positioned anchorage</td><td>$\leq d_f$</td><td colspan="2">14</td><td>18</td><td>22</td><td>26</td></tr>
<tr><td>push through anchorage</td><td>$\leq d_f$</td><td colspan="2">18</td><td>22</td><td>26</td><td>32</td></tr>
<tr><td colspan="2">Minimum thickness of concrete member</td><td>h_{min}</td><td>$h_0 + 30$</td><td colspan="4">$h_0 + 2d_0$</td></tr>
<tr><td colspan="2">Maximum installation torque</td><td>max T_{inst}</td><td>[Nm]</td><td colspan="2">40</td><td>60</td><td>120</td><td>150</td></tr>
</table>

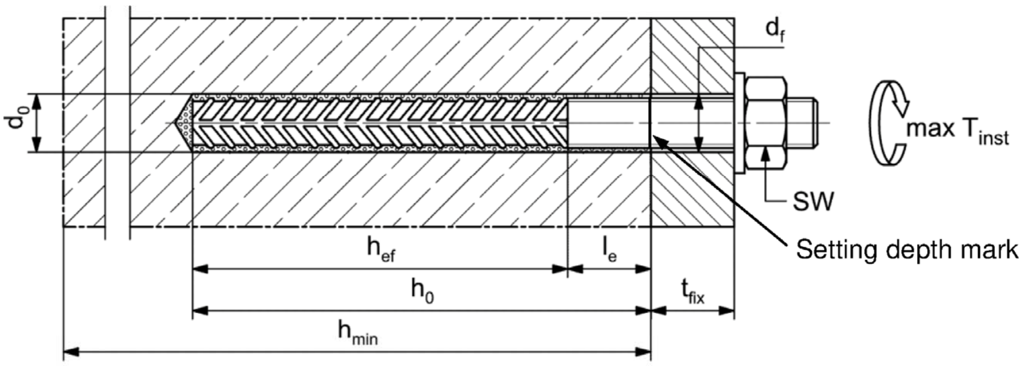
1) minimum spacing and minimum edge distance see Annex B 5
2) Both drill hole diameters can be used

fischer rebar anchor FRA



Marking frontal e.g: FRA (for stainless steel);
FRA HCR (for high corrosion resistant steel)

Installation conditions:



Figures not to scale

fischer injection system FIS V Plus

Intended use
Installation parameters rebar anchor FRA

Annex B 8

Table B9.1: Parameters of the **cleaning brush** BS (steel brush with steel bristles)

The size of the cleaning brush refers to the drill hole diameter

Nominal drill hole diameter	d_0	[mm]	8	10	12	14	16	18	20	24	25	28	30	35
Steel brush diameter	d_b		9	11	14	16	20		25	26	27	30	40	

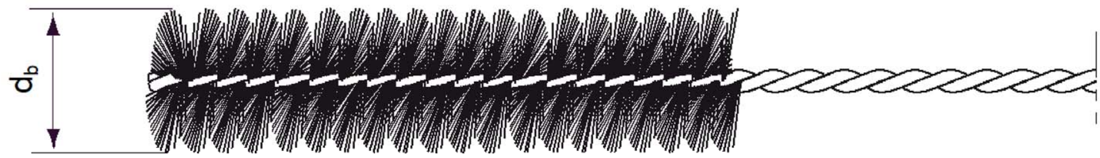


Table B9.2 **Maximum processing** time of the mortar and **minimum curing** time (During the curing time of the mortar the concrete temperature may not fall below the listed minimum temperature)

Temperature at anchoring base [°C]	Maximum processing time t_{work}			Minimum curing time 1) t_{cure}		
	FIS VW Plus High Speed	FIS V Plus	FIS VS Plus Low Speed	FIS VW Plus High Speed	FIS V Plus	FIS VS Plus Low Speed
-10 to -5 2)	-	-	-	12 h	-	-
> -5 to 0 2)	5 min	-	-	3 h	24 h	-
> 0 to 5 2)	5 min	13 min	-	3 h	3 h	6 h
> 5 to 10	3 min	9 min	20 min	50 min	90 min	3 h
> 10 to 20	1 min	5 min	10 min	30 min	60 min	2 h
> 20 to 30	-	4 min	6 min	-	45 min	60 min
> 30 to 40	-	2 min	4 min	-	35 min	30 min

1) In wet concrete or water filled holes the curing times must be doubled
2) Minimal cartridge temperature +5°C

fischer injection system FIS V Plus	**Annex B 9**
Intended use Cleaning brush (steel brush) Processing time and curing time	

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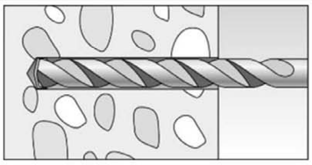
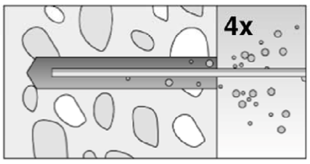
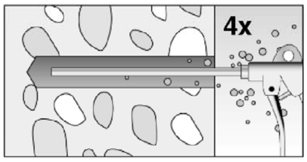
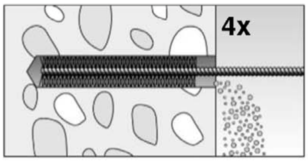
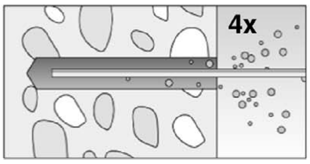
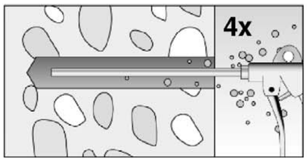
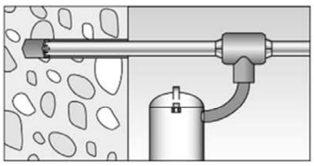
Installation instructions part 1

Drilling and cleaning the hole (hammer drilling with standard drill bit)

1		Drill the hole. Nominal drill hole diameter **d_0** and drill hole depth **h_0** see **tables B3.1, B6.1, B7.1, B8.1**		
2		Clean the drill hole: For $h_{ef} \leq 12d$ and $d_0 < 18$ mm blow out the hole four times by hand		For $h_{ef} > 12d$ and / or $d_0 \geq 18$ mm blow out the hole four times with oil-free compressed air ($p \geq 6$ bar)
3		Brush the drill hole four times. For drill hole diameter ≥ 30 mm use a power drill. For deep holes use an extension. Corresponding brushes see **table B9.1**		
4		Clean the drill hole: For $h_{ef} \leq 12d$ and $d_0 < 18$ mm blow out the hole four times by hand		For $h_{ef} > 12d$ and / or $d_0 \geq 18$ mm blow out the hole four times with oil-free compressed air ($p \geq 6$ bar)

Go to step 5

Drilling and cleaning the hole (hammer drilling with hollow drill bit)

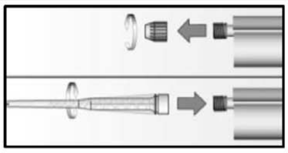
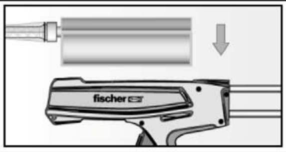
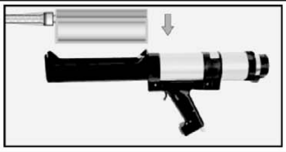
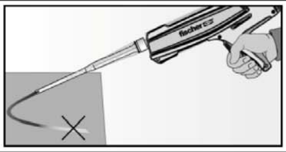
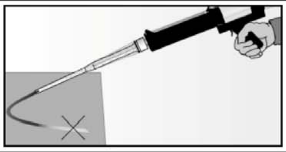
1		Check a suitable hollow drill (see **table B1.1**) for correct operation of the dust extraction
2		Use a suitable dust extraction system, e.g. fischer FVC 35 M or a comparable dust extraction system with equivalent performance data Drill the hole with hollow drill bit. The dust extraction system has to extract the drill dust nonstop during the drilling process and must be adjusted to maximum power. Nominal drill hole diameter **d_0** and drill hole depth **h_0** see **tables B3.1, B6.1, B7.1, B8.1**

Go to step 5

fischer injection system FIS V Plus	**Annex B 10**
Intended use Installation instructions part 1	

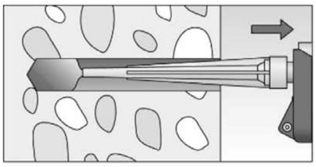
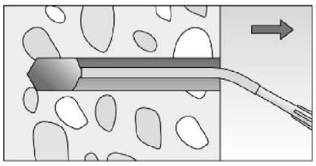
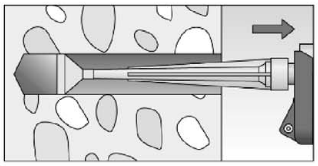
Installation instructions part 2

Preparing the cartridge

5		Remove the sealing cap Screw on the static mixer (the spiral in the static mixer must be clearly visible)	
6			Place the cartridge into the dispenser
7			Extrude approximately 10 cm of material out until the resin is evenly grey in colour. Do not use mortar that is not uniformly grey

Go to step 8

Injection of the mortar

8	 Fill approximately 2/3 of the drill hole with mortar. Always begin from the bottom of the hole and avoid bubbles	 For drill hole depth ≥ 150 mm use an extension tube	 For overhead installation, deep holes (h_0 > 250 mm) or drill hole diameter (d_0 ≥ 40 mm) use an injection adapter

Go to step 9

fischer injection system FIS V Plus

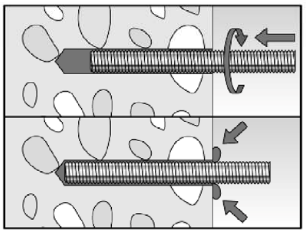
Intended use
Installation instructions part 2

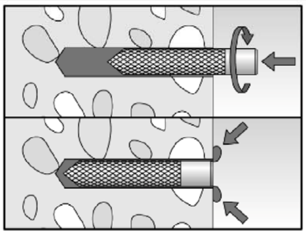
Annex B 11

Installation instructions part 3

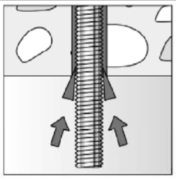
Installation of anchor rods or fischer internal threaded anchors RG MI

9

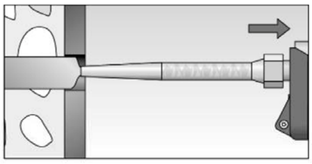




Only use clean and oil-free metal parts.
Mark the setting depth of the metal part. Push the anchor rod or fischer internal threaded RG MI anchor down to the bottom of the hole, turning it slightly while doing so.
After inserting the metal parts, excess mortar must be emerged around the anchor element.



For overhead installations support the metal part with wedges (e.g. fischer centering wedges) or fischer overhead clips.

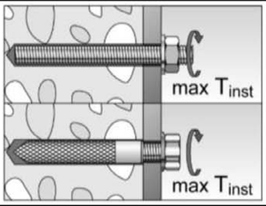


For push through installation fill the annular gap with mortar

10

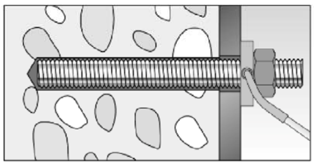
Wait for the specified curing time t_{cure}
see **table B9.2**

11



Mounting the fixture
max T_{inst} see **tables B3.1 and B6.1**

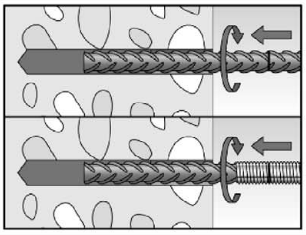
Option



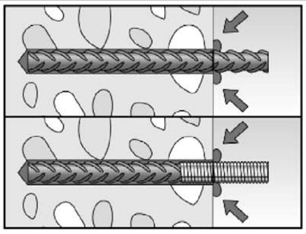
After the minimum curing time is reached, the gap between metal part and fixture (annular clearance) may be filled with mortar via the fischer filling disc. Compressive strength ≥ 50 N/mm² (e.g. fischer injection mortars FIS HB, FIS SB, FIS V, FIS V Plus, FIS EM Plus).
ATTENTION: Using fischer filling disc reduces t_{fix} (usable length of the anchor)

Installation reinforcing bars and fischer rebar anchor FRA

9



Only use clean and oil-free reinforcing bars or fischer FRA. Mark the setting depth. Turn while using force to push the reinforcement bar or the fischer FRA into the filled hole up to the setting depth mark.

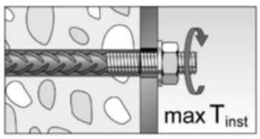


When the setting depth mark is reached, excess mortar must be emerged from the mouth of the drill hole.

10

Wait for the specified curing time t_{cure} see **table B9.2**

11



Mounting the fixture
max T_{inst}
see **table B8.1**

fischer injection system FIS V Plus	**Annex B 12**
Intended use Installation instructions part 3	

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Table C1.1: Characteristic values for steel failure under tension / shear load of **fischer anchor rods** and **standard threaded rods**

Anchor rod / standard threaded rod				**M6**	**M8**	**M10**	**M12**	**M16**	**M20**	**M24**	**M27**	**M30**
Bearing capacity under tension load, steel failure [3]												
Characteristic resistance $N_{Rk,s}$	Steel zinc plated	Property class 4.8	[kN]	8	15(13)	23(21)	33	63	98	141	184	224
		5.8		10	19(17)	29(27)	43	79	123	177	230	281
		8.8		16	29(27)	47(43)	68	126	196	282	368	449
	Stainless steel R and high corrosion resistant steel HCR	50		10	19	29	43	79	123	177	230	281
		70		14	26	41	59	110	172	247	322	393
		80		16	30	47	68	126	196	282	368	449
Partial factors [1]												
Partial factor $\gamma_{Ms,N}$	Steel zinc plated	Property class 4.8	[-]	1,50								
		5.8		1,50								
		8.8		1,50								
	Stainless steel R and high corrosion resistant steel HCR	50		2,86								
		70		1,50[2] / 1,87								
		80		1,60								
Bearing capacity under shear load, steel failure [3]												
without lever arm												
Characteristic resistance $V^0_{Rk,s}$	Steel zinc plated	Property class 4.8	[kN]	4	9(8)	14(13)	20	38	59	85	110	135
		5.8		6	11(10)	17(16)	25	47	74	106	138	168
		8.8		8	15(13)	23(21)	34	63	98	141	184	225
	Stainless steel R and high corrosion resistant steel HCR	50		5	9	15	21	39	61	89	115	141
		70		7	13	20	30	55	86	124	161	197
		80		8	15	23	34	63	98	141	184	225
Ductility factor		k_7	[-]	1,0								
with lever arm												
Characteristic resistance $M^0_{Rk,s}$	Steel zinc plated	Property class 4.8	[Nm]	6	15(13)	30(27)	52	133	259	448	665	899
		5.8		7	19(16)	37(33)	65	166	324	560	833	1123
		8.8		12	30(26)	60(53)	105	266	519	896	1333	1797
	Stainless steel R and high corrosion resistant steel HCR	50		7	19	37	65	166	324	560	833	1123
		70		10	26	52	92	232	454	784	1167	1573
		80		12	30	60	105	266	519	896	1333	1797
Partial factors [1]												
Partial factor $\gamma_{Ms,V}$	Steel zinc plated	Property class 4.8	[-]	1,25								
		5.8		1,25								
		8.8		1,25								
	Stainless steel R and high corrosion resistant steel HCR	50		2,38								
		70		1,25[2] / 1,56								
		80		1,33								

[1] In absence of other national regulations
[2] Only admissible for high corrosion resist. steel HCR, with f_{yk} / f_{uk} ≥ 0,8 and A_5 > 12 % (e.g. fischer anchor rods)
[3] Values in brackets are valid for undersized threaded rods with smaller stress area A_s for hot dip galvanised standard threaded rods according to EN ISO 10684:2004+AC:2009

fischer injection system FIS V Plus

Performances
Characteristic values for steel failure under tension / shear load of fischer anchor rods and standard threaded rods

Annex C 1

Table C2.1: Characteristic values for steel failure under tension / shear load of **fischer internal threaded anchors RG MI**

fischer internal threaded anchors RG MI					M8	M10	M12	M16	M20
Bearing capacity under tension load, steel failure									
Charact. resistance with screw	$N_{Rk,s}$	Property class	5.8	[kN]	19	29	43	79	123
			8.8		29	47	68	108	179
		Property class 70	R		26	41	59	110	172
			HCR		26	41	59	110	172
Partial factors[1)]									
Partial factors	$\gamma_{Ms,N}$	Property class	5.8	[-]	1,50				
			8.8		1,50				
		Property class 70	R		1,87				
			HCR		1,87				
Bearing capacity under shear load, steel failure									
Without lever arm									
Charact. resistance with screw	$V^0_{Rk,s}$	Property class	5.8	[kN]	9,2	14,5	21,1	39,2	62,0
			8.8		14,6	23,2	33,7	54,0	90,0
		Property class 70	R		12,8	20,3	29,5	54,8	86,0
			HCR		12,8	20,3	29,5	54,8	86,0
Ductility factor			k_7	[-]	1,0				
With lever arm									
Charact. resistance with screw	$M^0_{Rk,s}$	Property class	5.8	[Nm]	20	39	68	173	337
			8.8		30	60	105	266	519
		Property class 70	R		26	52	92	232	454
			HCR		26	52	92	232	454
Partial factors[1)]									
Partial factors	$\gamma_{Ms,V}$	Property class	5.8	[-]	1,25				
			8.8		1,25				
		Property class 70	R		1,56				
			HCR		1,56				

[1)] In absence of other national regulations

fischer injection system FIS V Plus	Annex C 2
Performances Characteristic values for steel failure under tension / shear load of fischer internal threaded anchor RG MI	

Table C3.1: Characteristic values for steel failure under tension / shear load of **reinforcing bars**

Nominal diameter of the bar		ϕ	8	10	12	14	16	20	25	28
Bearing capacity under tension load, steel failure										
Characteristic resistance	$N_{Rk,s}$	[kN]	$A_s \cdot f_{uk}$ [1]							
Bearing capacity under shear load, steel failure										
Without lever arm										
Characteristic resistance	$V^0_{Rk,s}$	[kN]	$0{,}5 \cdot A_s \cdot f_{uk}$ [1]							
Ductility factor	k_7	[-]	1,0							
With lever arm										
Characteristic resistance	$M^0_{Rk,s}$	[Nm]	$1{,}2 \cdot W_{el} \cdot f_{uk}$ [1]							

[1] f_{uk} or f_{yk} respectively must be taken from the specifications of the reinforcing bar

Table C3.2: Characteristic values for steel failure under tension / shear load of **fischer rebar anchors FRA**

fischer rebar anchor FRA			M12	M16	M20	M24
Bearing capacity under tension load, steel failure						
Characteristic resistance	$N_{Rk,s}$	[kN]	63	111	173	270
Partial factor[1]						
Partial factor	$\gamma_{Ms,N}$	[-]	1,4			
Bearing capacity under shear load, steel failure						
Without lever arm						
Characteristic resistance	$V^0_{Rk,s}$	[kN]	30	55	86	124
Ductility factor	k_7	[-]	1,0			
With lever arm						
Characteristic resistance	$M^0_{Rk,s}$	[Nm]	92	233	454	785
Partial factor[1]						
Partial factor	$\gamma_{Ms,V}$	[-]	1,56			

[1] In absence of other national regulations

Table C4.1: Characteristic values for concrete failure under tension / shear load

Size				All sizes	
Tension load					
Installation factor		γ_{inst}	[-]	See annex C 5 to C 12 and C 17 to C18	
Factors for the compressive strength of concrete > C20/25					
Increasing factor for τ_{Rk}	C25/30	Ψ_c	[-]	1,05	
	C30/37			1,10	
	C35/45			1,15	
	C40/50			1,19	
	C45/55			1,22	
	C50/60			1,26	
Splitting failure					
Edge distance	$h / h_{ef} \geq 2{,}0$	$c_{cr,sp}$	[mm]	$1{,}0\ h_{ef}$	
	$2{,}0 > h / h_{ef} > 1{,}3$			$4{,}6\ h_{ef} - 1{,}8\ h$	
	$h / h_{ef} \leq 1{,}3$			$2{,}26\ h_{ef}$	
Spacing		$s_{cr,sp}$		$2\ c_{cr,sp}$	
Concrete failure					
Uncracked concrete		$k_{ucr,N}$	[-]	11,0	
Cracked concrete		$k_{cr,N}$		7,7	
Edge distance		$c_{cr,N}$	[mm]	$1{,}5\ h_{ef}$	
Spacing		$s_{cr,N}$		$2\ c_{cr,N}$	
Factors for sustained tension load					
Temperature range			[-]	50 °C / 80 °C	72 °C / 120 °C
Factor		Ψ^0_{sus}	[-]	0,76	0,78
Shear load					
Installation factor		γ_{inst}	[-]	1,0	
Concrete pry-out failure					
Factor for pry-out failure		k_8	[-]	2,0	
Concrete edge failure					
Effective length of fastener in shear loading		l_f	[mm]	for $d_{nom} \leq 24$ mm: min (h_{ef}; 12 d_{nom}) for $d_{nom} > 24$ mm: min (h_{ef}; 8 d_{nom}; 300 mm)	

Calculation diameters

Size			M6	M8	M10	M12	M16	M20	M24	M27	M30
fischer anchor rods and standard threaded rods	d_{nom}	[mm]	6	8	10	12	16	20	24	27	30
fischer internal threaded anchors RG MI	d_{nom}		-[1]	12	16	18	22	28	-[1]	-[1]	-[1]
fischer rebar anchor FRA	d_{nom}		-[1]	-[1]	-[1]	12	16	20	25	-[1]	-[1]

Size (nominal diameter of the bar)	ϕ		8	10	12	14	16	20	25	28
Reinforcing bar	d_{nom}	[mm]	8	10	12	14	16	20	25	28

[1] Size of anchor type not part of the assessment

Table C5.1: **Characteristic values** for combined pull-out and concrete failure for **fischer anchor rods** and **standard threaded rods** in hammer drilled holes; **uncracked or cracked concrete; working life 50 years**

Anchor rod / standard threaded rod				M6	M8	M10	M12	M16	M20	M24	M27	M30
Combined pullout and concrete cone failure												
Calculation diameter		d	[mm]	6	8	10	12	16	20	24	27	30
Uncracked concrete												
Characteristic bond resistance in uncracked concrete C20/25												
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)												
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,ucr}$	[N/mm²]	9,0	16,0	16,0	15,0	14,0	12,0	11,0	10,0	9,0
Temperature range	II: 72 °C / 120 °C	$\tau_{Rk,ucr}$	[N/mm²]	6,5	15,0	14,0	13,0	12,0	11,0	9,0	8,0	8,0
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)												
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,ucr}$	[N/mm²]	-[1]	-[1]	-[1]	9,5	8,5	8,0	7,5	7,0	7,0
Temperature range	II: 72 °C / 120 °C	$\tau_{Rk,ucr}$	[N/mm²]	-[1]	-[1]	-[1]	7,5	7,0	6,5	6,0	6,0	6,0
Installation factors												
Dry or wet concrete		γ_{inst}	[-]	1,0								
Water filled hole		γ_{inst}	[-]	-[1]	-[1]	-[1]	1,2					
Cracked concrete												
Characteristic bond resistance in cracked concrete C20/25												
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)												
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,cr}$	[N/mm²]	-[1]	5,5	6,0	6,5	6,0	5,5	5,0	5,0	4,5
Temperature range	II: 72 °C / 120 °C	$\tau_{Rk,cr}$	[N/mm²]	-[1]	4,5	5,0	6,0	5,5	5,0	4,5	4,0	4,0
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)												
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,cr}$	[N/mm²]	-[1]	-[1]	-[1]	5,0	5,0	4,5	4,0	3,5	3,5
Temperature range	II: 72 °C / 120 °C	$\tau_{Rk,cr}$	[N/mm²]	-[1]	-[1]	-[1]	4,0	4,0	4,0	3,5	3,0	3,0
Installation factors												
Dry or wet concrete		γ_{inst}	[-]	1,0								
Water filled hole		γ_{inst}	[-]	-[1]	-[1]	-[1]	1,2					

[1] No performance assessed

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Table C6.1: **Characteristic values** for combined pull-out and concrete failure for **fischer anchor rods** and **standard threaded rods** in hammer drilled holes; **uncracked or cracked concrete; working life 100 years**

Anchor rod / standard threaded rod			**M6**	**M8**	**M10**	**M12**	**M16**	**M20**	**M24**	**M27**	**M30**
Combined pullout and concrete cone failure											
Calculation diameter	d	[mm]	6	8	10	12	16	20	24	27	30
Uncracked concrete											
Characteristic bond resistance in uncracked concrete C20/25											
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)											
Temperature range I: 50 °C / 80 °C	$\tau_{Rk,100,ucr}$	[N/mm²]	-1)	16,0	16,0	15,0	14,0	12,0	11,0	10,0	9,0
Temperature range II: 72 °C / 120 °C	$\tau_{Rk,100,ucr}$	[N/mm²]	-1)	15,0	14,0	13,0	12,0	11,0	9,0	8,0	8,0
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)											
Temperature range I: 50 °C / 80 °C	$\tau_{Rk,100,ucr}$	[N/mm²]	-1)	-1)	-1)	9,5	8,5	8,0	7,5	7,0	7,0
Temperature range II: 72 °C / 120 °C	$\tau_{Rk,100,ucr}$	[N/mm²]	-1)	-1)	-1)	7,5	7,0	6,5	6,0	6,0	6,0
Installation factors											
Dry or wet concrete	γ_{inst}	[-]	1,0								
Water filled hole	γ_{inst}	[-]	-1)	-1)	-1)	1,2					
Cracked concrete											
Characteristic bond resistance in cracked concrete C20/25											
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)											
Temperature range I: 50 °C / 80 °C	$\tau_{Rk,100,cr}$	[N/mm²]	-1)	5,0	5,5	5,5	5,5	5,5	5,0	5,0	4,5
Temperature range II: 72 °C / 120 °C	$\tau_{Rk,100,cr}$	[N/mm²]	-1)	4,5	5,0	5,0	5,0	5,0	4,0	4,0	4,0
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)											
Temperature range I: 50 °C / 80 °C	$\tau_{Rk,100,cr}$	[N/mm²]	-1)	-1)	-1)	4,5	4,5	4,5	4,0	3,5	3,5
Temperature range II: 72 °C / 120 °C	$\tau_{Rk,100,cr}$	[N/mm²]	-1)	-1)	-1)	4,0	4,0	4,0	3,5	3,0	3,0
Installation factors											
Dry or wet concrete	γ_{inst}	[-]	1,0								
Water filled hole	γ_{inst}	[-]	-1)	-1)	-1)	1,2					

1) No performance assessed

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Performances
Characteristic values for combined pull-out and concrete failure for fischer anchor rod and standard threaded rods; working life 100 years

Annex C 6

Table C7.1: Characteristic values for combined pull-out and concrete failure for **fischer internal threaded anchors RG MI** in hammer drilled holes; **uncracked concrete; working life 50 years**

Internal threaded anchor RG MI				**M8**	**M10**	**M12**	**M16**	**M20**
Combined pullout and concrete cone failure								
Calculation diameter		d	[mm]	12	16	18	22	28
Uncracked concrete								
Characteristic bond resistance in uncracked concrete C20/25								
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)								
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,ucr}$	[N/mm²]	10,5	10,0	9,5	9,0	8,5
	II: 72 °C / 120 °C			9,0	8,0	8,0	7,5	7,0
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)								
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,ucr}$	[N/mm²]	10,0	9,0	9,0	8,5	8,0
	II: 72 °C / 120 °C			7,5	6,5	6,5	6,0	6,0
Installation factors								
Dry or wet concrete		γ_{inst}	[-]	1,0				
Water filled hole				1,2				

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Performances Charactersitic values for combined pull-out and concrete failure for fischer internal threaded anchors RG MI; working life 50 years	

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Table C8.1: **Characteristic values** for combined pull-out and concrete failure for **reinforcing bars** in hammer drilled holes; **uncracked or cracked concrete; working life 50 years**

Nominal diameter of the bar			**ϕ**	**8**	**10**	**12**	**14**	**16**	**20**	**25**	**28**
Combined pullout and concrete cone failure											
Calculation diameter		d	[mm]	8	10	12	14	16	20	25	28
Uncracked concrete											
Characteristic bond resistance in uncracked concrete C20/25											
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)											
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,ucr}$	[N/mm²]	11,0	11,0	11,0	10,0	10,0	9,5	9,0	8,5
	II: 72 °C / 120 °C			9,5	9,5	9,0	8,5	8,5	8,0	7,5	7,0
Installation factor											
Dry or wet concrete		γ_{inst}	[-]	1,0							
Cracked concrete											
Characteristic bond resistance in cracked concrete C20/25											
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)											
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,cr}$	[N/mm²]	-[1]	3,0	5,0	5,0	5,0	4,5	4,0	4,0
	II: 72 °C / 120 °C			-[1]	3,0	4,5	4,5	4,5	4,0	3,5	3,5
Installation factor											
Dry or wet concrete		γ_{inst}	[-]	1,0							

[1] No performance assessed

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Performances
Characteristic values for combined pull-out and concrete failure for reinforcing bars; working life 50 years

Annex C 8

Table C9.1: **Characteristic values** for combined pull-out and concrete failure for **fischer rebar anchors FRA** in hammer drilled holes; **uncracked or cracked concrete; working life 50 years**

fischer rebar anchor FRA				**M12**	**M16**	**M20**	**M24**
Combined pullout and concrete cone failure							
Calculation diameter		d	[mm]	12	16	20	25
Uncracked concrete							
Characteristic bond resistance in uncracked concrete C20/25							
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)							
Tem-perature range	I: 50 °C / 80 °C	$\tau_{Rk,ucr}$	[N/mm²]	11,0	10,0	9,5	9,5
	II: 72 °C / 120 °C			9,0	8,5	8,0	7,5
Installation factors							
Dry or wet concrete		γ_{inst}	[-]	1,0			
Cracked concrete							
Characteristic bond resistance in cracked concrete C20/25							
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)							
Tem-perature range	I: 50 °C / 80 °C	$\tau_{Rk,cr}$	[N/mm²]	5,0	5,0	4,5	4,0
	II: 72 °C / 120 °C			4,5	4,5	4,0	3,5
Installation factors							
Dry or wet concrete		γ_{inst}	[-]	1,0			

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Table C10.1: Displacements for anchor rods

Anchor rod		M6	M8	M10	M12	M16	M20	M24	M27	M30
Displacement-Factors for tension load[1]										
Uncracked concrete; Temperature range I, II										
$\delta_{N0\text{-Factor}}$	[mm/(N/mm²)]	0,09	0,09	0,09	0,10	0,10	0,10	0,10	0,11	0,12
$\delta_{N\infty\text{-Factor}}$		0,10	0,10	0,10	0,12	0,12	0,12	0,13	0,13	0,14
Cracked concrete; Temperature range I, II										
$\delta_{N0\text{-Factor}}$	[mm/(N/mm²)]	-[3]	0,12	0,12	0,12	0,13	0,13	0,13	0,14	0,15
$\delta_{N0\text{-Factor}}$		-[3]	0,25	0,27	0,30	0,30	0,30	0,35	0,35	0,40
Displacement-Factors for shear load[2]										
Uncracked or cracked concrete; Temperature range I, II										
$\delta_{V0\text{-Factor}}$	[mm/kN]	0,11	0,11	0,11	0,10	0,10	0,09	0,09	0,08	0,07
$\delta_{V\infty\text{-Factor}}$		0,12	0,12	0,12	0,11	0,11	0,10	0,10	0,09	0,09

[1] Calculation of effective displacement:

$\delta_{N0} = \delta_{N0\text{-Factor}} \cdot \tau_{Ed}$

$\delta_{N\infty} = \delta_{N\infty\text{-Factor}} \cdot \tau_{Ed}$

(τ_{Ed}: Design value of the applied tensile stress)

[2] Calculation of effective displacement:

$\delta_{V0} = \delta_{V0\text{-Factor}} \cdot V_{Ed}$

$\delta_{V\infty} = \delta_{V\infty\text{-Factor}} \cdot V_{Ed}$

(V_{Ed}: Design value of the applied shear force)

[3] No performance assessed

Table C10.2: Displacements for fischer internal threaded anchors RG MI

Internal threaded anchor RG MI		M8	M10	M12	M16	M20
Displacement-Factors for tension load[1]						
Uncracked concrete; Temperature range I, II						
$\delta_{N0\text{-Factor}}$	[mm/(N/mm²)]	0,10	0,11	0,12	0,13	0,14
$\delta_{N\infty\text{-Factor}}$		0,13	0,14	0,15	0,16	0,18
Displacement-Factors for shear load[2]						
Uncracked concrete; Temperature range I, II						
$\delta_{V0\text{-Factor}}$	[mm/kN]	0,12	0,12	0,12	0,12	0,12
$\delta_{V\infty\text{-Factor}}$		0,14	0,14	0,14	0,14	0,14

[1] Calculation of effective displacement:

$\delta_{N0} = \delta_{N0\text{-Factor}} \cdot \tau_{Ed}$

$\delta_{N\infty} = \delta_{N\infty\text{-Factor}} \cdot \tau_{Ed}$

(τ_{Ed}: Design value of the applied tensile stress)

[2] Calculation of effective displacement:

$\delta_{V0} = \delta_{V0\text{-Factor}} \cdot V_{Ed}$

$\delta_{V\infty} = \delta_{V\infty\text{-Factor}} \cdot V_{Ed}$

(V_{Ed}: Design value of the applied shear force)

Table C11.1: Displacements for reinforcing bars

Nominal diameter of the bar	ϕ	8	10	12	14	16	20	25	28
Displacement-Factors for tension load[1]									
Uncracked concrete; Temperature range I, II									
δ_{N0}-Factor	[mm/(N/mm²)]	0,09	0,09	0,10	0,10	0,10	0,10	0,10	0,11
$\delta_{N\infty}$-Factor	[mm/(N/mm²)]	0,10	0,10	0,12	0,12	0,12	0,12	0,13	0,13
Cracked concrete; Temperature range I, II									
δ_{N0}-Factor	[mm/(N/mm²)]	-[3]	0,12	0,13	0,13	0,13	0,13	0,13	0,14
$\delta_{N\infty}$-Factor	[mm/(N/mm²)]	-[3]	0,27	0,30	0,30	0,30	0,30	0,35	0,37
Displacement-Factors for shear load[2]									
Uncracked or cracked concrete; Temperature range I, II									
δ_{V0}-Factor	[mm/kN]	0,11	0,11	0,10	0,10	0,10	0,09	0,09	0,08
$\delta_{V\infty}$-Factor	[mm/kN]	0,12	0,12	0,11	0,11	0,11	0,10	0,10	0,09

[1] Calculation of effective displacement:

$\delta_{N0} = \delta_{N0\text{-Factor}} \cdot \tau_{Ed}$

$\delta_{N\infty} = \delta_{N\infty\text{-Factor}} \cdot \tau_{Ed}$

(τ_{Ed}: Design value of the applied tensile stress)

[2] Calculation of effective displacement:

$\delta_{V0} = \delta_{V0\text{-Factor}} \cdot V_{Ed}$

$\delta_{V\infty} = \delta_{V\infty\text{-Factor}} \cdot V_{Ed}$

(V_{Ed}: Design value of the applied shear force)

[3] No performance assessed

Table C11.2: Displacements for fischer rebar anchors FRA

fischer rebar anchor FRA		M12	M16	M20	M24
Displacement-Factors for tension load[1]					
Uncracked concrete; Temperature range I, II					
δ_{N0}-Factor	[mm/(N/mm²)]	0,10	0,10	0,10	0,10
$\delta_{N\infty}$-Factor	[mm/(N/mm²)]	0,12	0,12	0,12	0,13
Cracked concrete; Temperature range I, II					
δ_{N0}-Factor	[mm/(N/mm²)]	0,12	0,13	0,13	0,13
$\delta_{N\infty}$-Factor	[mm/(N/mm²)]	0,30	0,30	0,30	0,35
Displacement-Factors for shear load[2]					
Uncracked or cracked concrete; Temperature range I, II					
δ_{V0}-Factor	[mm/kN]	0,10	0,10	0,09	0,09
$\delta_{V\infty}$-Factor	[mm/kN]	0,11	0,11	0,10	0,10

[1] Calculation of effective displacement:

$\delta_{N0} = \delta_{N0\text{-Factor}} \cdot \tau_{Ed}$

$\delta_{N\infty} = \delta_{N\infty\text{-Factor}} \cdot \tau_{Ed}$

(τ_{Ed}: Design value of the applied tensile stress)

[2] Calculation of effective displacement:

$\delta_{V0} = \delta_{V0\text{-Factor}} \cdot V_{Ed}$

$\delta_{V\infty} = \delta_{V\infty\text{-Factor}} \cdot V_{Ed}$

(V_{Ed}: Design value of the applied shear force)

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Table C12.1: Characteristic values for steel failure under tension / shear load of fischer anchor rods and standard threaded rods under seismic action performance category C1 or C2

Anchor rod / standard threaded rod					M10	M12	M16	M20	M24	M27	M30
Bearing capacity under tension load, steel failure[1]											
fischer anchor rods and standard threaded rods, performance category C1 [2]											
Characteristic resistance $N_{Rk,s,C1}$	Steel zinc plated	Property class	5.8	[kN]	29(27)	43	79	123	177	230	281
			8.8		47(43)	68	126	196	282	368	449
	Stainless steel R and high corrosion resistant steel HCR		50		29	43	79	123	177	230	281
			70		41	59	110	172	247	322	393
			80		47	68	126	196	282	368	449
fischer anchor rods, performance category C2 [2]											
Characteristic reistance $N_{Rk,s,C2}$	Steel zinc plated	Property class	5.8	[kN]	-[4]	39	72	108	-[4]	-[4]	-[4]
			8.8		-[4]	61	116	173	-[4]	-[4]	-[4]
	Stainless steel R and high corrosion resistant steel HCR		50		-[4]	39	72	108	-[4]	-[4]	-[4]
			70		-[4]	53	101	152	-[4]	-[4]	-[4]
			80		-[4]	61	116	173	-[4]	-[4]	-[4]
Bearing capacity under shear load, steel failure without lever arm[1]											
fischer anchor rods, performance category C1 [2]											
Characteristic resistance $V_{Rk,s,\ C1}$	Steel zinc plated	Property class	5.8	[kN]	17(16)	25	47	74	106	138	168
			8.8		23(21)	34	63	98	141	184	225
	Stainless steel R and high corrosion resistant steel HCR		50		15	21	39	61	89	115	141
			70		20	30	55	86	124	161	197
			80		23	34	63	98	141	184	225
Standard threaded rods, performance category C1 [2]											
Characteristic resistance $V_{Rk,s,\ C1}$	Steel zinc plated	Property class	5.8	[kN]	12(11)	17	33	52	74	97	118
			8.8		16(14)	24	44	69	99	129	158
	Stainless steel R and high corrosion resistant steel HCR		50		11	15	27	43	62	81	99
			70		14	21	39	60	87	113	138
			80		16	24	44	69	99	129	158
fischer anchor rods, performance category C2											
Characteristic resistance $V_{Rk,s,\ C2}$	Steel zinc plated	Property class	5.8	[kN]	-[4]	14	27	43	-[4]	-[4]	-[4]
			8.8		-[4]	22	44	69	-[4]	-[4]	-[4]
	Stainless steel R and high corrosion resistant steel HCR		50		-[4]	14	27	43	-[4]	-[4]	-[4]
			70		-[4]	20	39	60	-[4]	-[4]	-[4]
			80		-[4]	22	44	69	-[4]	-[4]	-[4]
Factor for the annular gap		α_{gap}		[-]	0,5 (1,0)[3]						

[1] Partial factors for performance category C1 or C2 see table C13.1;
for fischer anchor rods FIS A / RGM the factor for steel ductility is 1,0
[2] Values in brackets are valid for undersized threaded rods with smaller stress area A_s and for hot dip galvanised standard threaded rods according to EN ISO 10684:2004+AC:2009.
[3] Values in brackets are valid for filled annular gaps between the anchor rod and the through-hole in the attachment. It is necessary to use the fischer filling disc according to Annex A 5
[4] No performance assessed

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Performances
Characteristic values for steel failure under tension / shear load for fischer anchor rods and standard threaded rods under seismic action (performance category C1 / C2)

Annex C 12

Table C13.1: Partial factors for fischer anchor rods, standard threaded rods under seismic action performance category **C1 or C2**

Anchor rod / standard threaded rod				**M10**	**M12**	**M16**	**M20**	**M24**	**M27**	**M30**
Tension load, steel failure[1)]										
Partial factor $\gamma_{Ms,N}$	Steel zinc plated	Property class	5.8	[-]	1,50					
			8.8		1,50					
	Stainless steel R and high corrosion resistant steel HCR		50		2,86					
			70		1,50[2)] / 1,87					
			80		1,60					
Shear load, steel failure[1)]										
Partial factor $\gamma_{Ms,V}$	Steel zinc plated	Property class	5.8	[-]	1,25					
			8.8		1,25					
	Stainless steel R and high corrosion resistant steel HCR		50		2,38					
			70		1,25[2)] / 1,56					
			80		1,33					

[1)] In absence of other national regulations
[2)] Only admissible for high corrosion resistant steel HCR, with f_{yk} / f_{uk} ≥ 0,8 and A_5 > 12 % (e.g. fischer anchor rods)

Table C14.1: Characteristic values for combined pull-out and concrete failure for **fischer anchor rods** and **standard threaded rods** in hammer drilled holes under seismic action performance category **C1, working life 50 and 100 years**

Anchor rod / standard threaded rod				**M10**	**M12**	**M16**	**M20**	**M24**	**M27**	**M30**
Characteristic bond resistance, combined pullout and concrete cone failure										
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)										
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,C1}$	[N/mm²]	4,5	5,5	5,5	5,5	4,5	4,0	4,0
	II: 72 °C / 120 °C			4,0	4,5	4,5	4,5	4,0	3,5	3,5
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)										
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,C1}$	[N/mm²]	-[1)]	5,0	5,0	4,5	4,0	3,5	3,5
	II: 72 °C / 120 °C			-[1)]	4,0	4,0	4,0	3,5	3,0	3,0
Installation factors										
Dry or wet concrete		γ_{inst}	[-]	1,0						
Water filled hole				-[1)]	1,2					

[1)] No performance assessed

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Performances
Characteristic values under seismic action (performance category C1) for fischer anchor rods and standard threaded rods, working life 50 and 100 years

Annex C 14

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Table C15.1: Characteristic values for combined pull-out and concrete failure for **fischer anchor rods** in hammer drilled holes under seismic action performance category **C2; working life 50 and 100 years**

Anchor rod / standard threaded rod				M12	M16	M20
Characteristic bond resistance, combined pullout and concrete cone failure						
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)						
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,C2}$	[N/mm²]	1,5	1,3	2,1
	II: 72 °C / 120 °C			1,3	1,2	1,9
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)						
Temperature range	I: 50 °C / 80 °C	$\tau_{Rk,C2}$	[N/mm²]	1,3	1,1	1,8
	II: 72 °C / 120 °C			1,1	1,0	1,6
Displacement-Factors for tension load[1)]						
$\delta_{N,C2\ (DLS)}$-Factor			[mm/(N/mm²)]	0,20	0,13	0,21
$\delta_{N,C2\ (ULS)}$-Factor				0,38	0,18	0,24
Displacement-Factors for shear load[2)]						
$\delta_{V,C2\ (DLS)}$-Factor			[mm/kN]	0,18	0,10	0,07
$\delta_{V,C2\ (ULS)}$-Factor				0,25	0,14	0,11

1) Calculation of effective displacement:

$\delta_{N,C2\ (DLS)} = \delta_{N,C2\ (DLS)\text{-Factor}} \cdot \tau_{Ed}$

$\delta_{N,C2\ (ULS)} = \delta_{N,C2\ (ULS)\text{-Factor}} \cdot \tau_{Ed}$

(τ_{Ed}: Design value of the applied tensile stress)

2) Calculation of effective displacement:

$\delta_{V,C2\ (DLS)} = \delta_{V,C2\ (DLS)\text{-Factor}} \cdot V_{Ed}$

$\delta_{V,C2\ (ULS)} = \delta_{V,C2\ (ULS)\text{-Factor}} \cdot V_{Ed}$

(V_{Ed}: Design value of the applied shear force)

3) No performance assessed

Product Testing

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Eurofins Product Testing A/S
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VOC EMISSION TEST REPORT
CDPH

19 September 2022

1 Sample Information

Sample name	FIS V Plus
Batch no.	E52A55
Stated production date	01/06/2022
Product type	Chemical anchor
Sample reception	19/07/2022

2 Brief Evaluation of the Results

Regulation or protocol	Conclusion	Version of regulation or protocol
CDPH §		CDPH/EHLB/Standard Method V1.2. (January 2017)
- Classroom scenario	Fail	
- Office scenario	Pass	

Full details based on the testing and direct comparison with limit values are available in the following pages
Regarding pass/fail decision rule please see appendix
§*See section 4.4 on deviations*

Laura Hartung Sørensen
Analytical Service Manager

Rasmus Verdier
Analytical Service Manager

The results are only valid for the tested sample(s).
This report may only be copied or reprinted in its entity.

Table of contents

3 Applied Test Methods

3.1 General Test References

Regulation, protocol or standard	Version	Reporting limit VOC [µg/m³]	Calculation of TVOC	Combined uncertainty¤ [RSD(%)]
EN 16516	2017 + A1:2020	5	Toluene equivalents	22%
ISO 16000 -3 -6 -9 -11	2006-2021 depending on part	2	Toluene equivalents	22%
ASTM D5116-10	2010	-	-	-
CDPH	CDPH/EHLB/Standard Method V1.2. (January 2017)	2	Toluene equivalents	22%

3.2 Specific Laboratory Sampling and Analyses

Procedure	External Method	Internal SOP	Quantification limit / sampling volume	Analytical principle	Uncertainty¤ [RSD(%)]
Sample preparation	ISO 16000-11:2006, EN 16516:2017+A1:2020, CDPH:2017	71M549810	-	-	-
Emission chamber testing	ISO 16000-9:2006, EN 16516:2017+A1:2020	71M549811	-	Chamber and air control	-
Sampling of VOC	ISO 16000-6:2021, EN 16516:2017+A1:2020	71M549812	5 L	Tenax TA	-
Analysis of VOC	ISO 16000-6:2021, EN 16516:2017+A1:2020	71M542808B	1 µg/m³	ATD-GC/MS	10%
Sampling of aldehydes	ISO 16000-3:2011, EN 16516:2017+A1:2020	71M549812	35 L	DNPH	-
Analysis of aldehydes	ISO 16000-3:2011, EN 16516:2017+A1:2020	71M548400	3-6 µg/m³	HPLC-UV	10%
Sampling on Charcoal tubes	ISO 16200-1:2001	71M549812	60 L	Charcoal	-
Analysis of Charcoal tubes *	ISO-16200-1:2001	71M546081	20 µg/m³	Headspace-GC/MS	10%

4 Test Parameters, Sample Preparation and Deviations

4.1 VOC Emission Chamber Test Parameters

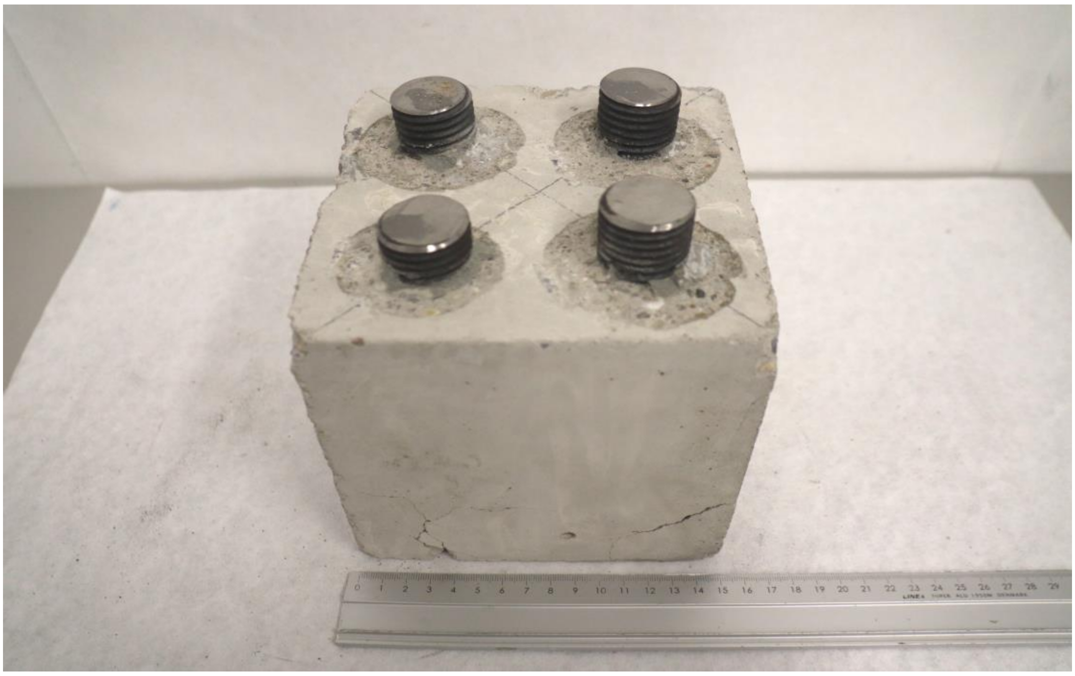
Parameter	Value	Parameter	Value
Chamber volume, V[L]	119	Preconditioning period	-
Air Change rate, n[h^{-1}]	1.0	Chamber test period	29/08/2022 - 12/09/2022
Area specific ventilation rate, q [m/h or $m^3/m^2/h$]	143	Analytical test period	29/08/2022 - 16/09/2022
Relative humidity of supply air, RH [%]	50 ± 3	Loading factor [m^2/m^3]	0.007 §
Temperature of supply air, T [°C]	23 ± 1	Test scenario	Very small area §

§See section 4.4 on deviations

4.2 Preparation of the Test Specimen

Approximately 2/3 of each drill hole in the concrete block was filled with mortar and metal anchor rods were pushed into the drill holes. The excess mortar was removed after curing.

4.3 Picture of Sample



4.4 Deviations from Referenced Protocols and Regulations

The loading factor was less than the lowest factor of 0.3 m^2/m^3 that CDPH method specifies for testing; CDPH method does not specify a clear loading factor in any model room. Instead, the loading factor as specified in EN 16516 was applied both during testing and for calculation of the air concentration in office and classroom.

5 Results

5.1 VOC Emission Test Results after 11 Days§

	CAS No.	Specific Conc. [µg/m³]	Specific SER [µg/(m²·h)]	Toluene eq. [µg/m³]	Toluene SER [µg/(m²·h)]
TVOC (C5-C17)tol. eq.				190	27000
Aldehydes					
Formaldehyde	50-00-0	3.0	430		
Acetaldehyde	75-07-0	< 3	< 500		

§See section 4.4 on deviations

5.2 VOC Emission Test Results after 12 Days§

	CAS No.	Specific Conc. [µg/m³]	Specific SER [µg/(m²·h)]	Toluene eq. [µg/m³]	Toluene SER [µg/(m²·h)]
TVOC (C5-C17)tol. eq.				200	29000
Aldehydes					
Formaldehyde	50-00-0	< 3	< 500		
Acetaldehyde	75-07-0	< 3	< 500		

§See section 4.4 on deviations

5.3 VOC Emission Test Results after 14 Days§

	CAS No.	Retention time [min]	ID-Cat	SER [µg/(m²·h)]	Classroom Conc. [µg/m³]	Office Conc. [µg/m³]	½ CREL [µg/m³]
VOC (C5-C17)							
Benzene	71-43-2	2.51	1	460	3.9	0.5	1.5
1-Chloro-2-propanol *	127-00-4	2.85	2	380	3.2	0.4	
Methylmethacrylate	80-62-6	3.09	1	380	3.2	0.4	
1-Methoxy-2-propyl acetate	108-65-6	6.38	1	2100	18	2.2	
n-Propylbenzene	103-65-1	7.89	1	800	6.8	0.8	
1,3,5-Trimethylbenzene	108-67-8	8.13	1	1100	9.4	1.1	
1,2,4-Trimethylbenzene	95-63-6	8.53	1	3900	33	4.0	
Methyl methacrylate *	27813-02-1	9.22	2	3000	26	3.1	
Not identified *		9.29	4	1700	15	1.7	
Not identified *		14.57	4	2100	18	2.2	
TVOC (C5-C17)tol. eq.				23000	200	24	

Aldehydes							
Formaldehyde	50-00-0		1	< 500	< 5	< 1	9
Acetaldehyde	75-07-0		1	< 500	< 5	< 1	70

§See section 4.4 on deviations

6 Summary and Evaluation of the Results

6.1 Comparison with Limit Values of CDPH

Parameter	Test after 14 days			
	CAS No. Single compounds	Concentration in Classroom [µg/m³]	Concentration in Office Room [µg/m³]	½ CREL [µg/m³]
TVOC (C5-C17)tol. eq.	-	200	24	-
Single compounds (with defined CREL values)				
Benzene	71-43-2	3.9	0.5	1.5
Formaldehyde	50-00-0	< 5	< 1	≤ 9
Acetaldehyde	75-07-0	< 5	< 1	≤ 70

6.1.1 Conversion of Emission Rates to CDPH Reference Room Concentrations

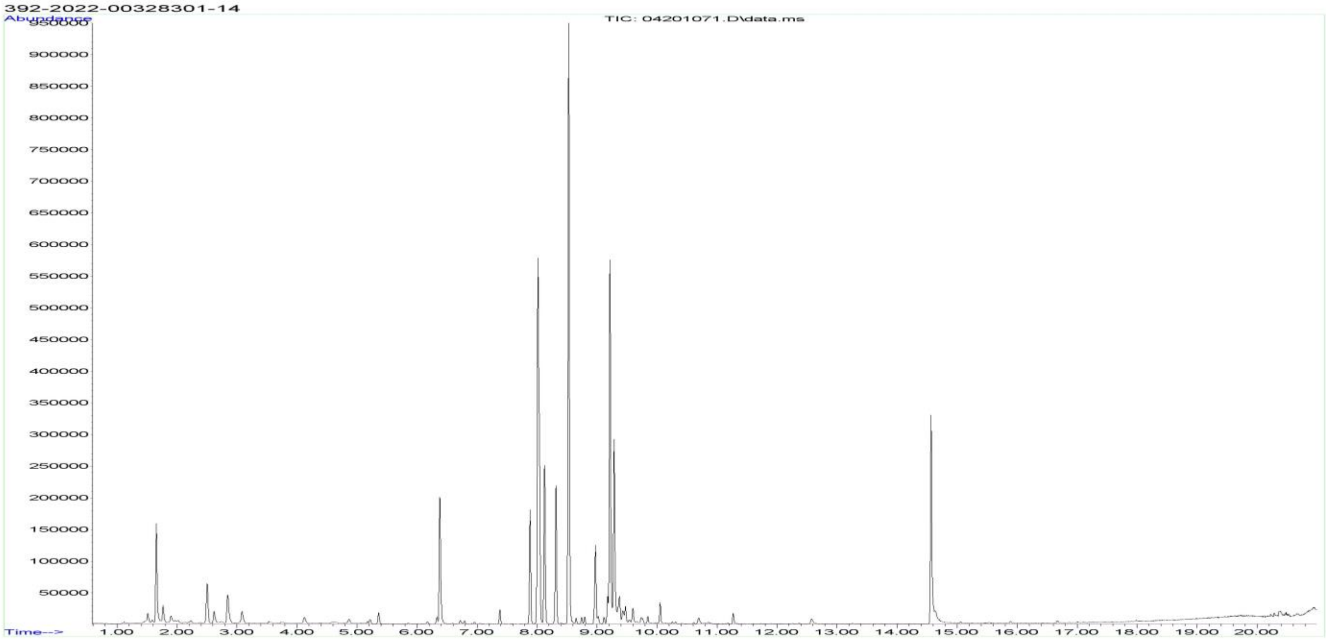
The CDPH method requires calculation of the measured emission rates into concentrations in given reference rooms. The equation and parameters figured below have been applied to calculate the concentrations in an office room or a classroom as required in the CDPH. The area used in the calculation varies depending on the expected usage of the product and therefore several entries can be found. Small and Very Small areas are not provided within the CDPH but are adapted from definitions given in EN 16516 and ISO 16000-9.

$$C_{Calculated} = \frac{SER_A \cdot A}{n \cdot V}$$

		Classroom parameters	Office Room parameters
SER	Area specific emission rate, µg/(m²h)	As tested	As tested
n	Air change, h^{-1}	0.82	0.68
V	Volume of reference room, m³	231	30.6
A	Floor area, m²	89.2	11.1
	Walls area, m²	94.3	33.4
	Ceiling and Wall, m²	183.8	N/A
	Door and Millwork, m²	1.89	1.89
	Desk or Chair, units	27	1
	Very Small areas, m²	1.62	0.021
	Small areas, m²	11.55	1.53

7 Appendices

7.1 Chromatogram of VOC Emissions after 14 Days



7.2 Chain of Custody

Chain of Custody

Name of the product: FIS V Plus	**Type of product:** Injection mortar
Model / Program / Series: FIS V Plus 360 S	**Batch N°.:** E52A55
Article N°.: Misc. 558752	**Date of batch production:** 30.06.2022
Name of the manufacturer at the place of sampling (address / stamp): fischerwerke GmbH & Co. KG	**Manufacturer** (if deviating from company's name at the place of sampling):
Sample collector (Name, company, telephone): Yvonne Herbstritt, fischerwerke, +49 7666 902 2901	**Signature of sample collector:**

Sample is taken from	x the ongoing production ☐ stocks		**Date of sampling:** 01.07.2022
Number of Samples	2 cartridges		**Time: 13:00**
Where had the product been stored prior to sampling?	x Production ☐ Store ☐ Miscellaneous	**How had the product been stored prior to sampling?**	☐ open ☐ in the stack x wrapped up
	Place of storage: Production line, bureau		Packing material: PP cartridge, cardboard box

Further links in chain of custody (Name, function, company, telephone)	**Signature**
Further links in chain of custody (Name, function, company, telephone)	**Signature**
Sample sender (Name, company, telephone): Yvonne Herbstritt, fischerwerke, +49 7666 902 2901	**Signature of sample sender:** ~~Yvonne Herbstritt, fischerwerke, +49 7666 902 2901~~
Date and time of sending:	**Shipment details/Carrier:**

Where had the product sample been stored prior to sending?	☐ Production ☐ Store x Miscellaneous	**How had the product sample been stored prior to sending?**	☐ open ☐ in the stack ☐ wrapped up
	Place of storage: bureau		Packing material: PP cartridge, cardboard box

Laboratory receiving details (date, condition of package and sample, assigned lab no.):	
Receptionist, Eurofins Product Testing A/S:	**Signature of receptionist:**

7.3 How to Understand the Results

7.3.1 Acronyms Used in the Report

<	Means less than
>	Means bigger than
*	Not a part of our accreditation
¤	Please see section regarding uncertainty in the Appendices
§	Deviation from method. Please see deviation section
a	The method is not optimal for very volatile compounds. For these substances smaller results and a higher measurement uncertainty cannot be ruled out
b	The component originates from the substrate and is thus removed
c	The results have been corrected by the emission from the substrate
d	Very polar organic compounds are not suitable for reliable quantification using Tenax TA adsorbent and HP-5ms GC column. A high degree of uncertainty must be expected
e	The component may be overestimated due to contribution from the system
SER	Specific Emission Rate

7.3.2 Explanation of ID Category

Categories of Identity:

1: Identified by comparison with a mass spectrum obtained from library and supported by other information and quantified through specific calibration.

2: Identified by comparison with a mass spectrum obtained from library and supported by other information. Quantified as toluene equivalent.

3: Identified with a lower match by comparison with a mass spectrum obtained from a library. Quantified as toluene equivalent.

4: Not identified, quantified as toluene equivalent.

7.4 Description of VOC Emission Test

7.4.1 Test Chamber

The test chamber is made of stainless steel. A multi-step air clean-up is performed before loading the chamber, and a blank check of the empty chamber is performed.
The chamber operation parameters are as described in the test method section. (EN 16516, ISO 16000-9, internal method no.: 71M549811).

7.4.2 Expression of the Test Results

All test results are calculated as specific emission rate, and as extrapolated air concentration in the European Reference Room (EN 16516, AgBB, EMICODE, M1 and Indoor Air Comfort).

7.4.3 Testing of Carcinogenic VOCs

The emission of carcinogens (EU Categories C1A and C1B, as per European law) is tested by drawing sample air from the test chamber outlet through Tenax TA tubes after the specified duration of storage in the ventilated test chamber. Analysis is performed by ATD-GC/MS (automated thermal desorption coupled with gas chromatography and mass spectroscopy using 30 m HP-5 (slightly polar) column with 0.25 mm ID and 0.25 µm film, Agilent) (EN 16516, ISO 16000-6, internal methods no.: 71M549812 / 71M542808B).

All identified carcinogenic VOCs are listed; if a carcinogenic VOC is not listed then it has not been detected. Quantification is performed using the TIC signal and authentic response factors, or the relative response factors relative to toluene for the individual compounds.

This test only covers substances that can be adsorbed on Tenax TA and can be thermally desorbed. If other emissions occur, then these substances cannot be detected (or with limited reliability only).

7.4.4 Testing of VOC

The emissions of volatile organic compounds are tested by drawing sample air from the test chamber outlet through Tenax TA tubes after the specified duration of storage in the ventilated test chamber. Analysis is performed by ATD-GC/MS using HP-5 column (30 m, 0.25mm ID, 0.25µm film).
This test only covers substances which can be adsorbed on Tenax TA and can be thermally desorbed. If emissions of substances outside these specifications occur then these substances cannot be detected (or with limited reliability only).

7.4.5 Testing of Aldehydes

The presence of aldehydes is tested by drawing air samples from the test chamber outlet through DNPH-coated silicagel tubes after the specified duration of storage in the ventilated test chamber. Analysis is performed by solvent desorption and subsequently by HPLC and UV-/diode array detection.

The absence of formaldehyde and other aldehydes is stated if UV detector response at the specific wavelength is lacking at the specific retention time in the chromatogram. Otherwise it is checked whether the reporting limit is exceeded. In this case the identity is finally checked by comparing full scan sample UV spectra with full scan standard UV spectra.

7.4.6 Testing of Charcoal tubes

The presence of low boiling VOC is tested by drawing air samples from the test chamber outlet through charcoal tubes after the specified duration of storage in the ventilated test chamber. Analysis is performed by solvent desorption and subsequently by HS-GC/MS using a stabilwax column. This test only covers substances which has a CREL value and are not possible to sample on Tenax tubes.

7.5 Quality Assurance

Before loading the test chamber, a blank check of the empty chamber is performed and compliance with background concentrations in accordance with EN 16516 / ISO 16000-9 is determined.

Air sampling at the chamber outlet and subsequent analysis is performed in duplicate. Relative humidity, temperature and air change rate in the chambers is logged every 5 minutes and checked daily. A double determination is performed on random samples at a regular interval and results are registered in a control chart to ensure the uncertainty and reproducibility of the method.

The stability of the analytical system is checked by a general function test of device and column, and by use of control charts for monitoring the response of individual substances prior to each analytical sequence.

7.6 Accreditation

The testing methods described above are accredited on line with EN ISO/IEC 17025 by DANAK (no. 522). This accreditation is valid worldwide due to mutual approvals of the national accreditation bodies (ILAC/IAF, see also www.eurofins.com/galten.aspx#accreditation).

Not all parameters are covered by this accreditation. The accreditation does not cover parameters marked with an asterisk (*), however analysis of these parameters is conducted at the same level of quality as for the accredited parameters.

7.7 Uncertainty of the Test Method

The relative standard deviation of the overall analysis is 22%. The expanded uncertainty Um equals 2 x RSD. For further information please visit www.eurofins.dk/uncertainty.

7.8 Decision Rules

Eurofins Product Testing A/S, declare statement of conformity based on the “Binary Statement for Simple Acceptance Rule” described in ILAC’s “Guidelines on decision Rules and Statements of Conformity” ILAC-G8:09/2019.

This means that results above the detection limit are always reported with two significant digits. Results are evaluated with the same number of significant digits as the corresponding limit values, and conformity is based on results being less than or equal to limit values.

For limit values with more than two significant digits, the third digit will be used to confirm whether a result is below or equal to the limit value. It will always be indicated in the evaluation table if this expanded evaluation is performed.

For further information please visit www.eurofins.dk/product-testing/om-os/beslutningsregler/

7.9 Version History

Report date	Report number	Modification
19/09/2022	392-2022-00328301_H_EN	Current version

Product Testing

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www.eurofins.com/VOC-testing

VOC TEST REPORT
VOC Content

1 April 2022

1 Sample Information

Sample name	FIS V Plus / FIS RC II / FIS VL
Sample no.	392-2022-00113501
Stated production date	18-02-2022
Batch No.	E99A01
Sample reception	10/03/2022

2 Brief Evaluation of the Results

Regulation or protocol	Conclusion	Version of regulation or protocol
SCAQMD Rule 1168	Pass	October 2017
LEED v4.1 (VOC Content)	Pass	

Full details based on the testing and direct comparison with limit values are available in the following pages
Regarding pass/fail decision rule please see appendix

Janne R Norup

Janne Rothmann Norup
Analytical Service Manager

The results are only valid for the tested sample(s).
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3 Applied Test Methods

3.1 General Test References

Regulation, protocol or standard	Scope	Version
SCAQMD Rule 1168	Adhesive and sealant applications	October 2017

3.2 Specific Laboratory Sampling and Analyses

Test	Regulation, protocol or standard	Version	Internal SOP	Limit of detection [g/L]	Uncertainty Um¤ %
Solids Content	ASTM D2369	2020	71 M 544830	1	10
VOC	ASTM D2369	2020	71 M 544830	1	10

3.3 Preparation of the Test Specimen

The sample was homogenised and applied directly onto the test dish.

4 Results

4.1 VOC content

	Remarks on the test results	Results	Unit
Density	Supplied by the customer	1.75	g/mL
Water content	Supplied by the customer	44	g/L
Exempt compounds *	Assumed to be 0	0	% (w/w)
Solids Content	Tested by the lab	94.7	% (w/w)
VOC content (less water)	Calculated based on the results above	51	g/L

4.2 Comparison with Limit Values of VOC Content (less Water)

Parameter	Results [g/L]	Product type	Regulation or protocol	VOC limit [g/L]
VOC content	51	Multipurpose construction adhesive	SCAQMD Rule 1168	75

5 Deviations from the referenced protocol

The density and water content was supplied by the customer and not tested by the laboratory.

6 Appendices

6.1 How to Understand the Results

6.1.1 Acronyms Used in the Report

< Means less than

> Means bigger than

* Not a part of our accreditation

¤ Please see section regarding uncertainty in the Appendices

1 Analysed by another Eurofins laboratory

6.2 Description of VOC Content Test

6.2.1 Testing of VOC

Volatile content of the sample was determined gravimetrically by heating to 110 °C in 60 minutes. Multicomponent products are mixed according to the manufacturer's instructions and allowed to cure before heating.

The result is the average of two replicates. The result was calculated as:

$$VOC = \frac{([g\ All\ Volatiles] - [g\ Water] - [g\ Exempt\ Compounds])}{([liter\ Material] - [liter\ Water] - [liter\ Exempt\ Compounds])}$$

6.3 Uncertainty of the Test Method

Um(%): The expanded uncertainty Um is equal to 2 x RSD%.

6.4 Decision Rules

Eurofins Product Testing A/S, declare statement of conformity based on the “Binary Statement for Simple Acceptance Rule” described in ILAC’s “Guidelines on decision Rules and Statements of Conformity” ILAC-G8:09/2019.

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For further information please visit www.eurofins.dk/product-testing/om-os/beslutningsregler/

6.5 Version History

Report date	Report number	Modification
01/04/2022	392-2022-00113501_XG_EN	Current version